SAP University Competence Center
Chair for Information Systems and Business Process Management
TUM School of Computation, Information, and Technology
Technical University of Munich

Advanced Practical Course – Next Generation Data Center Operations for SAP Enterprise Software (IN2128, IN2106, IN212810)

Containers

Simon Fuchs, Maximilian Niedermeier, Noah Kim, Leon Kist

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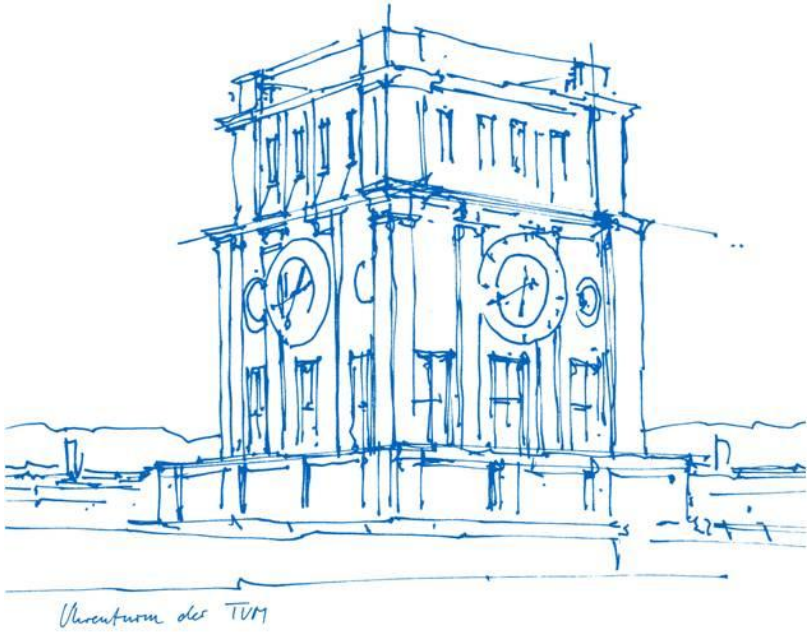

Container Orchestration

Introduction

Outline

- Containers in General
  - Definition
  - Container vs VM
- Container Environment
  - Container Registries
  - Building and Running Containers
  - Docker & Podman
- Container Orchestration → **How are containers scaled?**
  - Hyperscalers & Cloud
  - Kubernetes

Introduction

Group exercise:

- Form groups of 3-5 people → Your table
- As a group: Think about scaling (10 minutes)
  - Come up with 2 **everyday concepts** (outside of computing) that are scaled to a large/massive degree
  - Answer:
    - **What** is the concept?
    - **Why** does it need to be scaled?
    - **How** is it scaled, i.e., what mechanism is used? (Horizontal vs. Vertical)
- Present your results

Introduction

Group exercise:

- Example: Competitive Football (Soccer)
  - **What** is the concept?
    - Competitive Soccer
  - **Why** is it scaled?
    - Millions of people (in Germany) play → cannot hold one big tournament/game
  - **How** is it scaled, i.e., what mechanism is used?
    - Multi-tier league system (up to 13) with promotion/demotion, from nation-wide down to regionally locked
    - Horizontal scaling (many small- to mid-sized leagues instead of one large one)

Introduction

We have moved from **manual** container creation to **tools** for building, running, and monitoring

However:

- Tools (e.g., Docker & Podman) still require direct creation and management (by the user)
- Feasible for small to medium amount of containers
- **Does not scale!**

Introduction

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However:

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- Feasible for small to medium amount of containers
- **Does not scale!**

Solution → container orchestration tools

Hyperscaler & The Cloud

Who needs that many containers? And Why?

Hyperscaler & The Cloud

Who needs that many containers? And Why?

Regular company or datacenter:

- Dedicated (on-premise) servers
- Even if large number of applications exist → managed directly by teams

Problem: requires know-how, hardware expensive, long setup time, dedicated personal needed, …

Hyperscaler & The Cloud

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Solution → the cloud

Hyperscaler & The Cloud

What is the **cloud**?

- Applications are hosted externally
- Accessed over the internet
- (Often) Large dedicated datacenters in strategic locations
- Typically charge regular fees or depending on usage

Hyperscaler & The Cloud

Benefits (for users):

- No know-how required (for hardware and hosting) & no hardware costs
- Fast/immediate setup
- Managed externally → no personnel required
- Accessible for small companies (even in emerging markets) or private users
- Scaling

Hyperscaler & The Cloud

Problems (for users):

- Dependent on internet access
- Security and trust issues
- Dependency on providers
- Centralized hosting → single point of failure

Hyperscaler & The Cloud

Typical Services:

- Infrastructure as a service (IaaS) → IT infrastructure (e.g., computing, storage, …)
- Platform as a service (PaaS) → Tools and services to deploy applications (e.g., OS, runtime env., …)
- Software as a Service (SaaS) → Fully functional applications (e.g., ERP, CRM, …)

Hyperscaler & The Cloud

Who runs the cloud?

Hyperscaler & The Cloud

Who runs the cloud?

IBM Cloud

Sources: see references

Hyperscaler & The Cloud

Enterprise Datacenter vs Hyperscaler (cloud provider)

Enterprise Datacenter:

- Defined scope:
  - Number of applications
  - Nature of applications

- Highly Customized
- Limited number of applications
- Industry specific compliance
- Company specific bandwidth requirements
- Company specific redundancy requirements

Hyperscaler:

- Undefined scope:
  - (Unknown) Number of applications
  - (Unknown) Nature of applications

- High degree of **flexibility** required
- Millions of concurrent applications
- Global compliance
- Extreme bandwidth requirements
- High redundancy

Hyperscaler & The Cloud

How can hyperscalers/cloud providers provide these constraints?

Hyperscaler & The Cloud

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- Containers → flexibility, isolation, efficiency, …
- Container orchestration tools → scaling, redundancy, …

Hyperscaler & The Cloud

How can hyperscalers/cloud providers provide these constraints?

- Containers → flexibility, isolation, efficiency, …
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→ Kubernetes

Source: https://github.com/kubernetes/kubernetes/blob/master/logo/logo.svg

Kubernetes | Agenda

Agenda:

- History of Kubernetes
- What is Kubernetes?
  - Design
  - Architecture
- Features

Kubernetes | History

Kubernetes:

- Successor to *Borg:*
  - Developed at Google
  - Unified container-management system
  - Both long-running services and batch jobs
  - Utilizes Linux container support → Google contributed code to the Linux kernel

Kubernetes | History

Kubernetes:

- Kubernetes = “helmsman” (ancient Greek)
- Pitched in 2013 by Craig McLuckie, Joe Beda and Brendan Burns (Google)
- Released in 2014 → Open source
- Now: **second largest** open-source project in the world (after Linux)
- Cloud Native Computing Foundation graduated Project

Kubernetes | History

Cloud Native Computing Foundation:

- Part of the Linux Foundation
- Provide: support, oversight, and direction to **cloud native projects**
- Large amount of industry partners
- Vast landscape

Kubernetes | History

Cloud Native Computing Foundation:

CLOUD NATIVE LANDSCAPE

Source: https://landscape.cncf.io/images/logo_header.svg

Kubernetes | Design

Kubernetes:

- “[..] system for automating **deployment**, **scaling**, and **management** of containerized applications”
- (Some) Features:
  - Automated rollouts and rollbacks
  - Service discovery and load balancing
  - Horizontal scaling, vertical scaling
  - Secret and configuration management

Kubernetes | Design

Kubernetes:

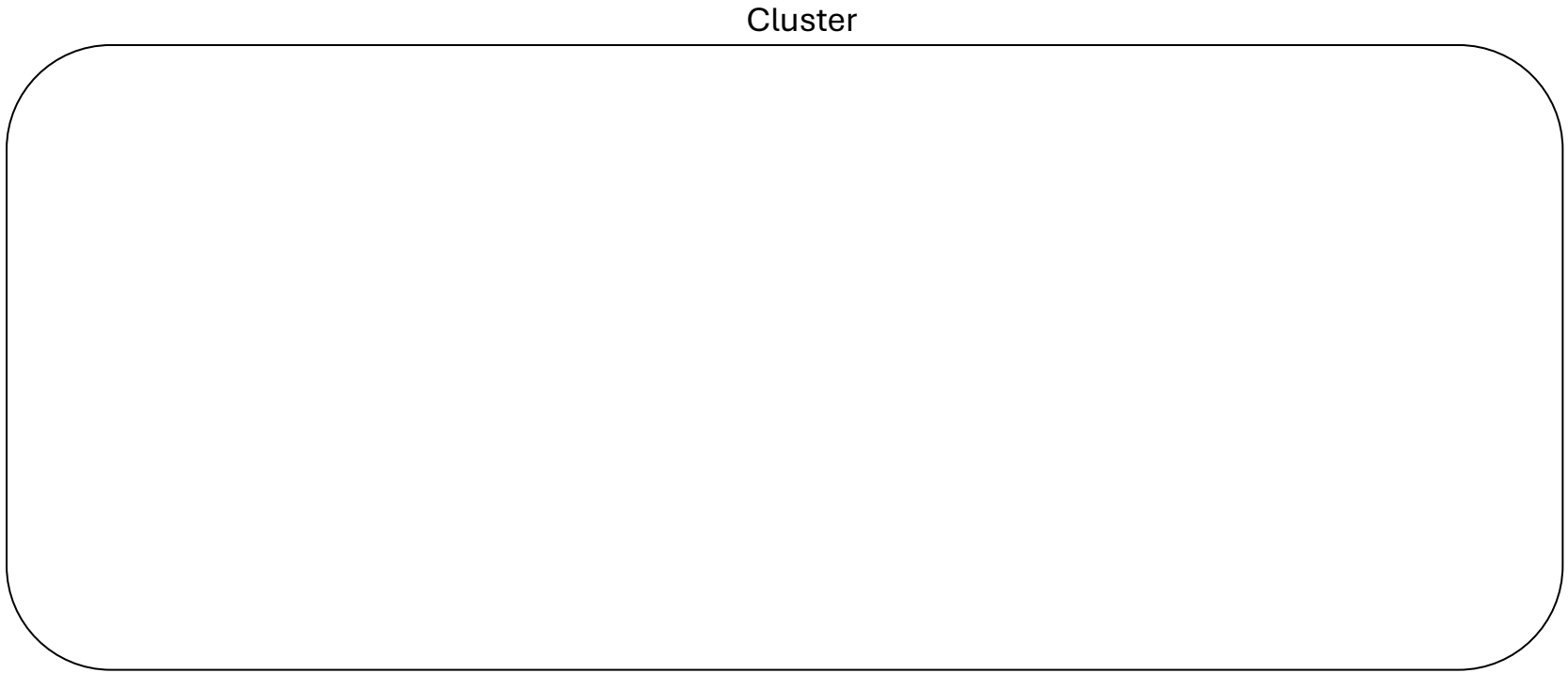
- Core **design principles**:
  - Immutability
  - Declarative configuration
  - Self-healing
  - Extensibility
  - Scalability

Kubernetes | Architecture

Kubernetes Cluster:

Kubernetes | Architecture

Kubernetes Cluster:



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Kubernetes | Architecture

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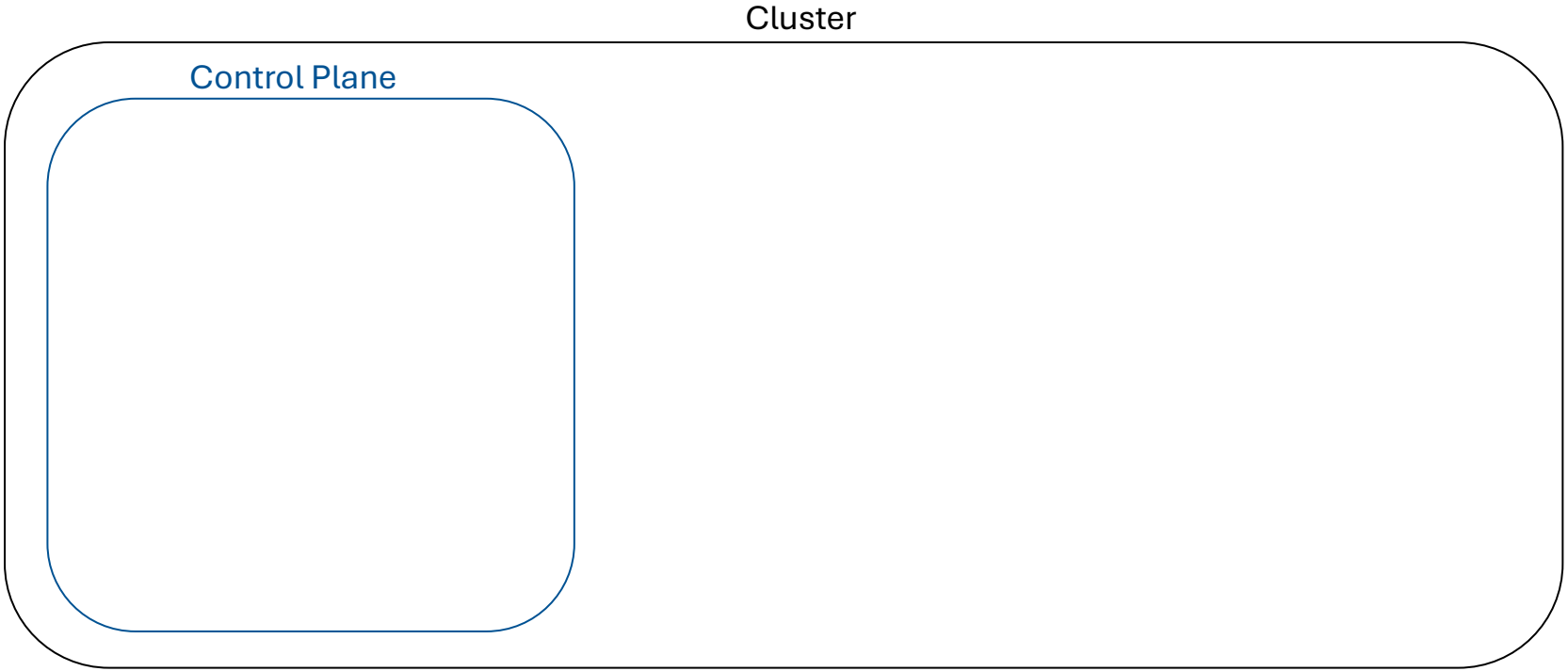

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Kubernetes | Architecture

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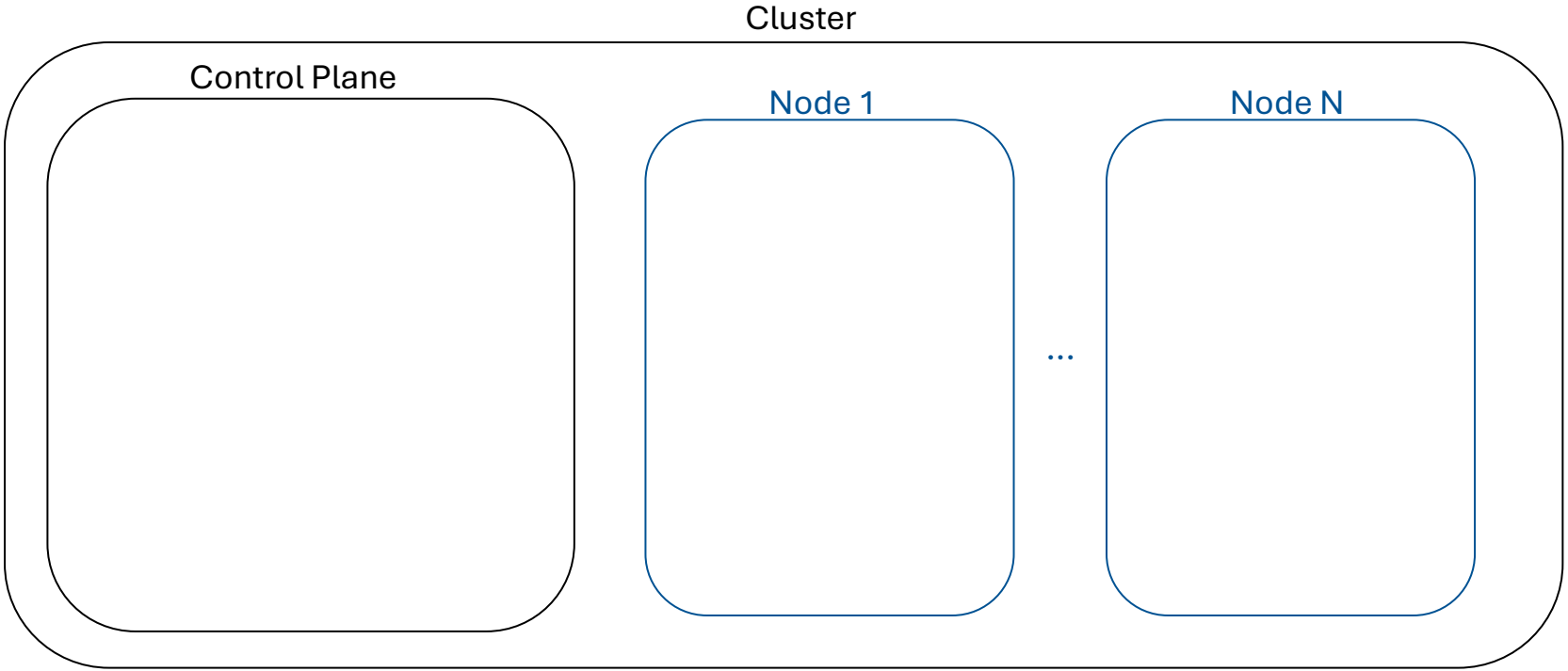


Based on: https://kubernetes.io/docs/concepts/architecture/

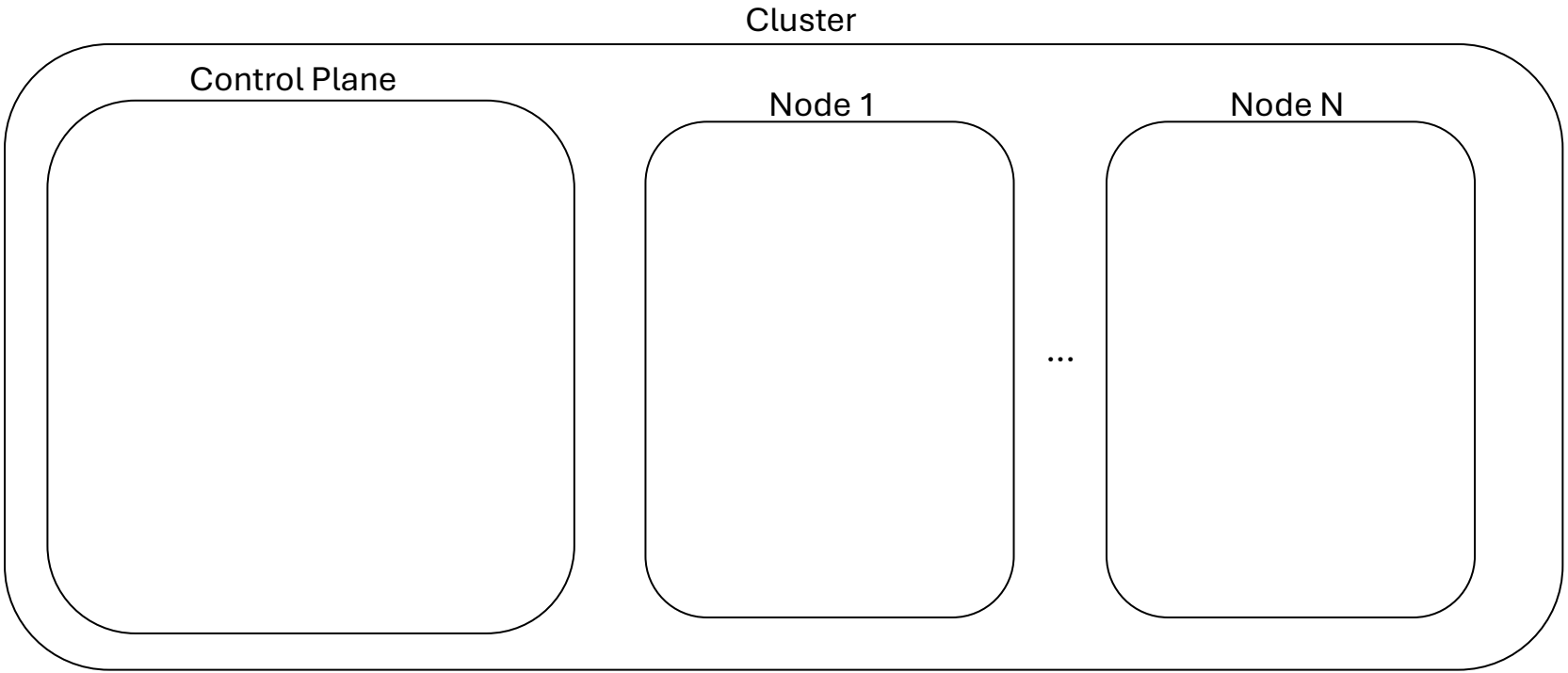
Kubernetes | Architecture

Kubernetes Cluster:

- **Nodes:**
  - Run containers/applications
- **Control Plane:**
  - Group of components that manages and orchestrates the cluster
  - Can run on any of the nodes in the cluster

Kubernetes | Architecture

Kubernetes Cluster:



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Kubernetes | Architecture

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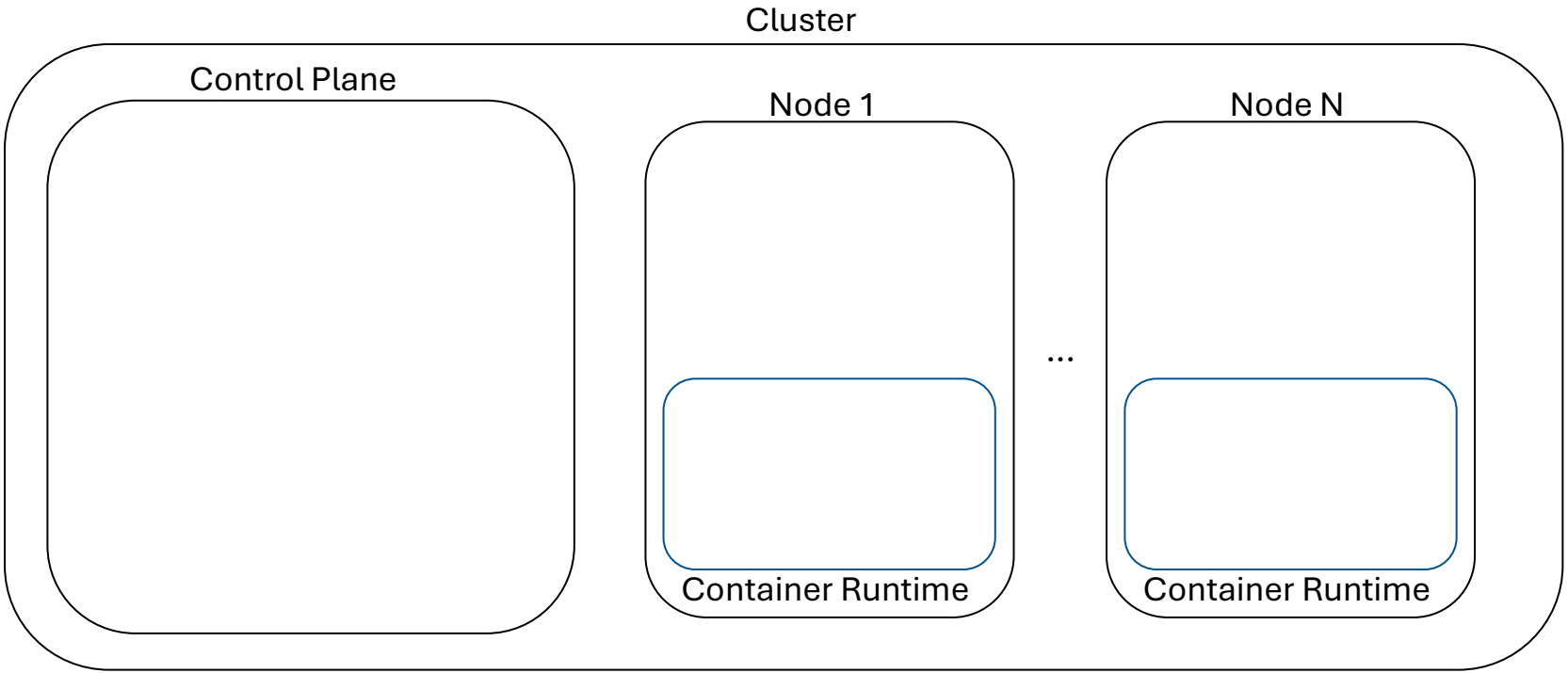

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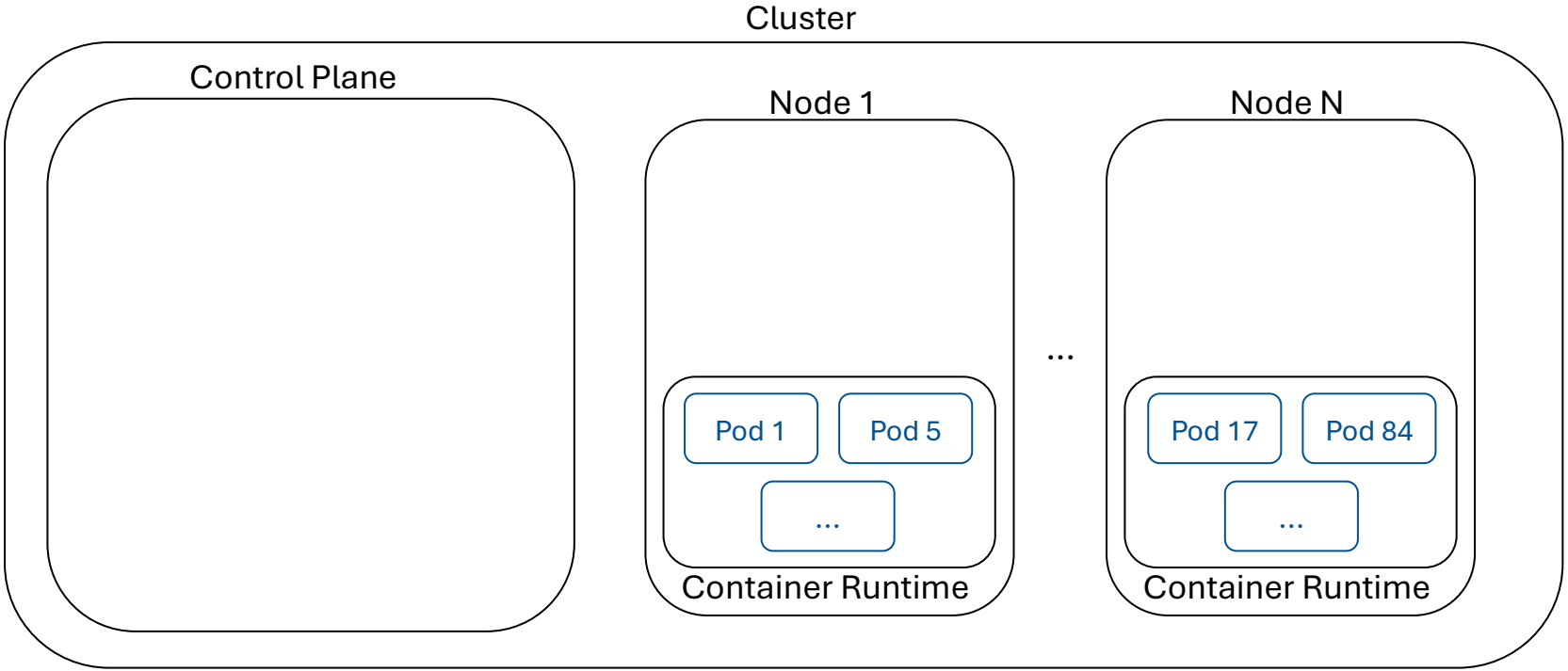
Kubernetes | Architecture

Container Runtime:

- Reminder → Responsible for:
  - Executing containers
  - Managing container lifecycle
- Has to implement Kubernetes Container Runtime Interface (CRI)
- Examples: containerd, CRI-O, Docker Engine

Kubernetes | Architecture

Kubernetes Cluster:



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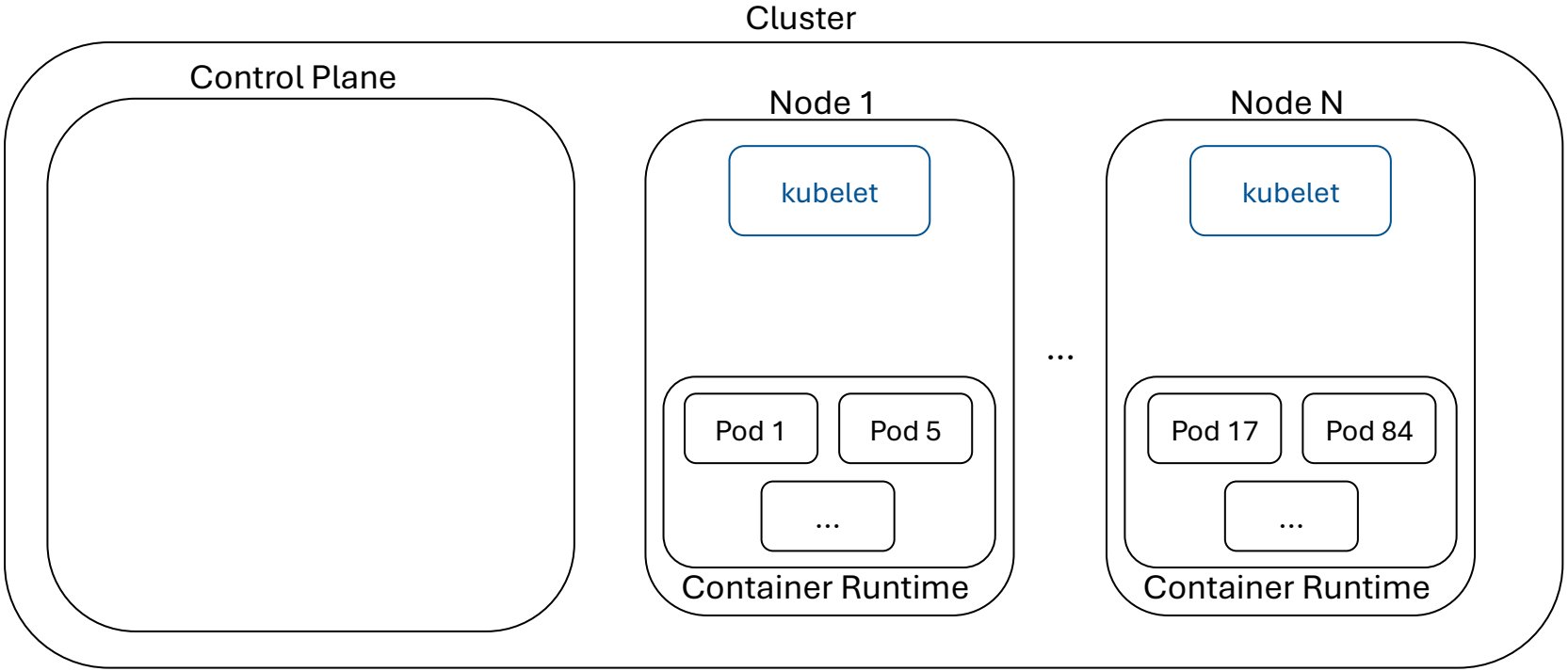
Kubernetes | Architecture

Pods:

- Smallest deployable unit of computation in Kubernetes
- Wrap a set of containers with shared namespaces and system volumes → always on the same node
- Usage:
  - Single-container pods: very common use case
  - Multi-container pods: containers that benefit from being on the same node or sharing resources

Kubernetes | Architecture

Kubernetes Cluster:



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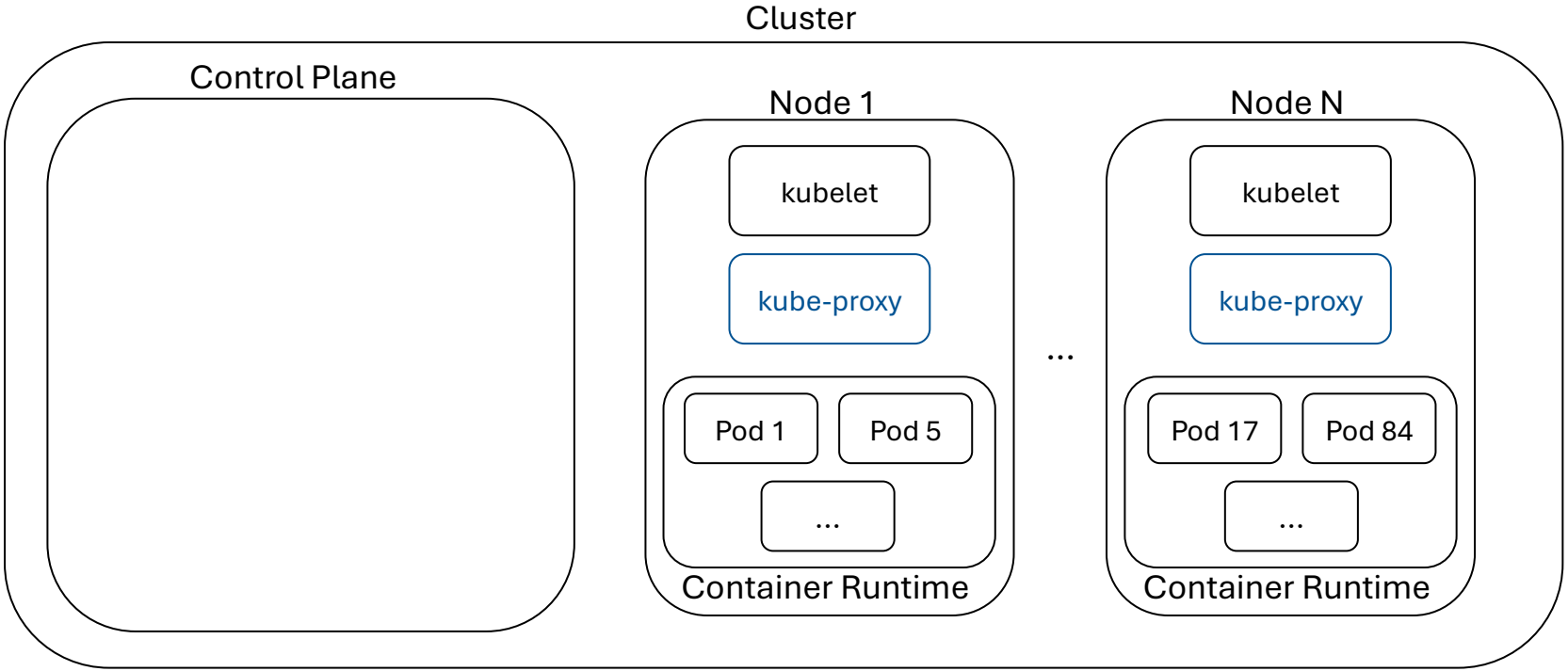
Kubernetes | Architecture

kubelet :

- Runs on each node
- Ensures that a specified set of containers are running and healthy
- Communicates with the control plane:
  - Registers the node with the cluster
  - Receives information about which containers to run (from the control plane)

Kubernetes | Architecture

Kubernetes Cluster:



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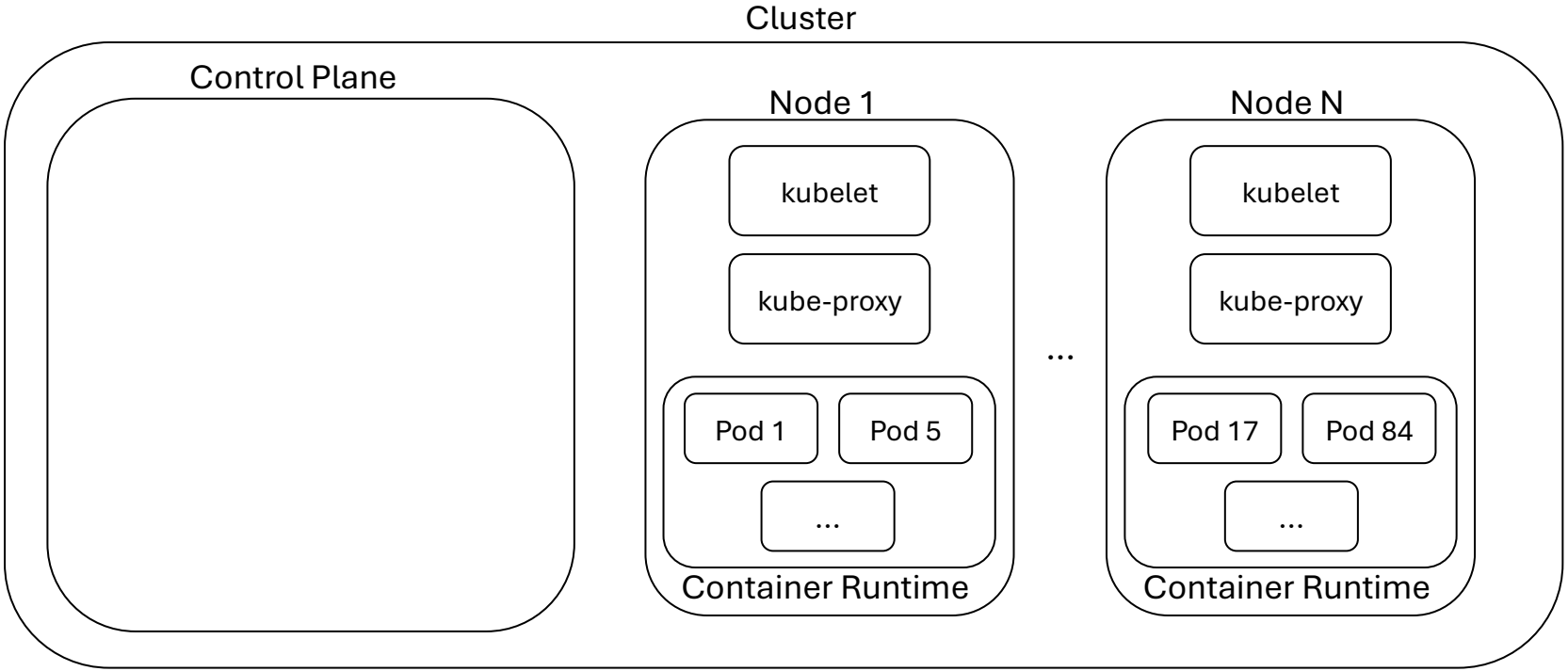
Kubernetes | Architecture

kube-proxy :

- (optionally) Runs on each node
- Network proxy
- Allows (network) communication with Pods from inside and outside the cluster
- Implements part of the functionality of Kubernetes Services & Service Discovery

Kubernetes | Architecture

Kubernetes Cluster:



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Kubernetes | Architecture

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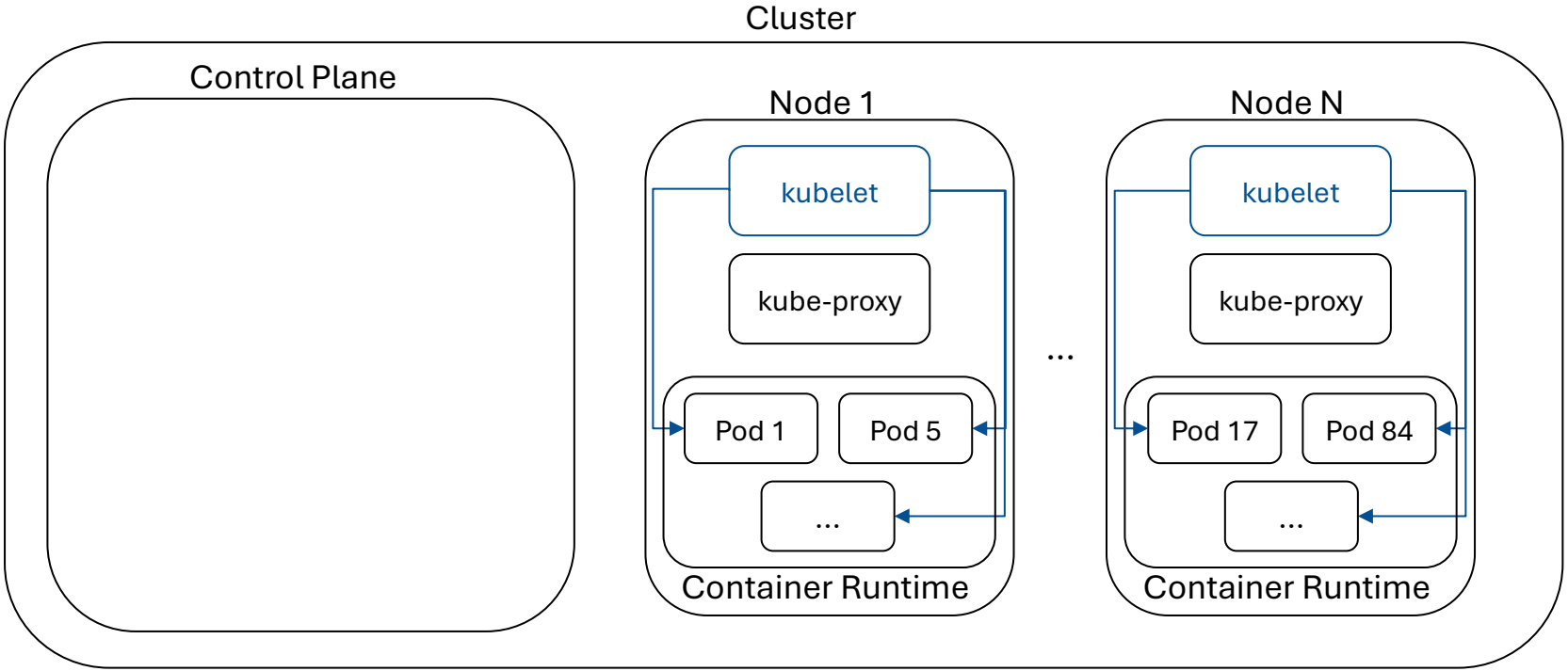

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Kubernetes | Architecture

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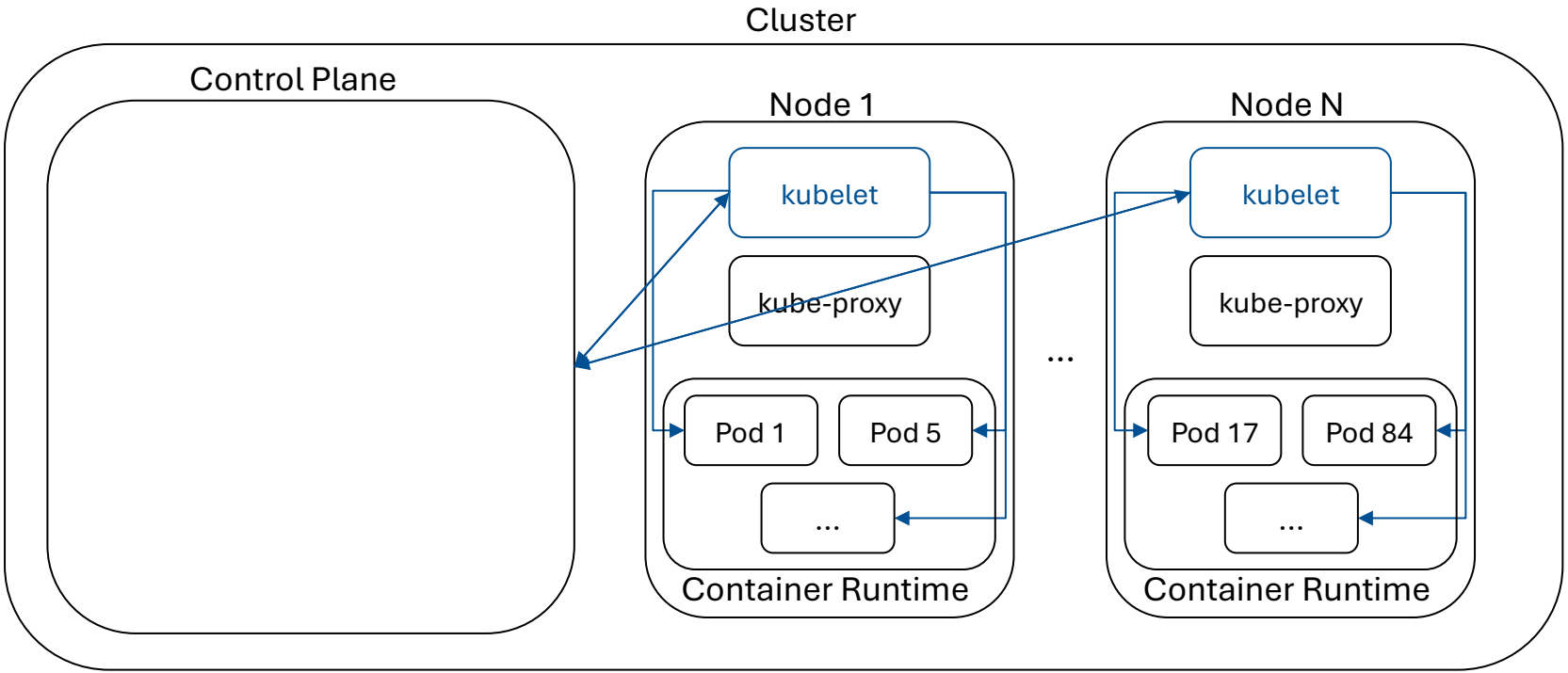

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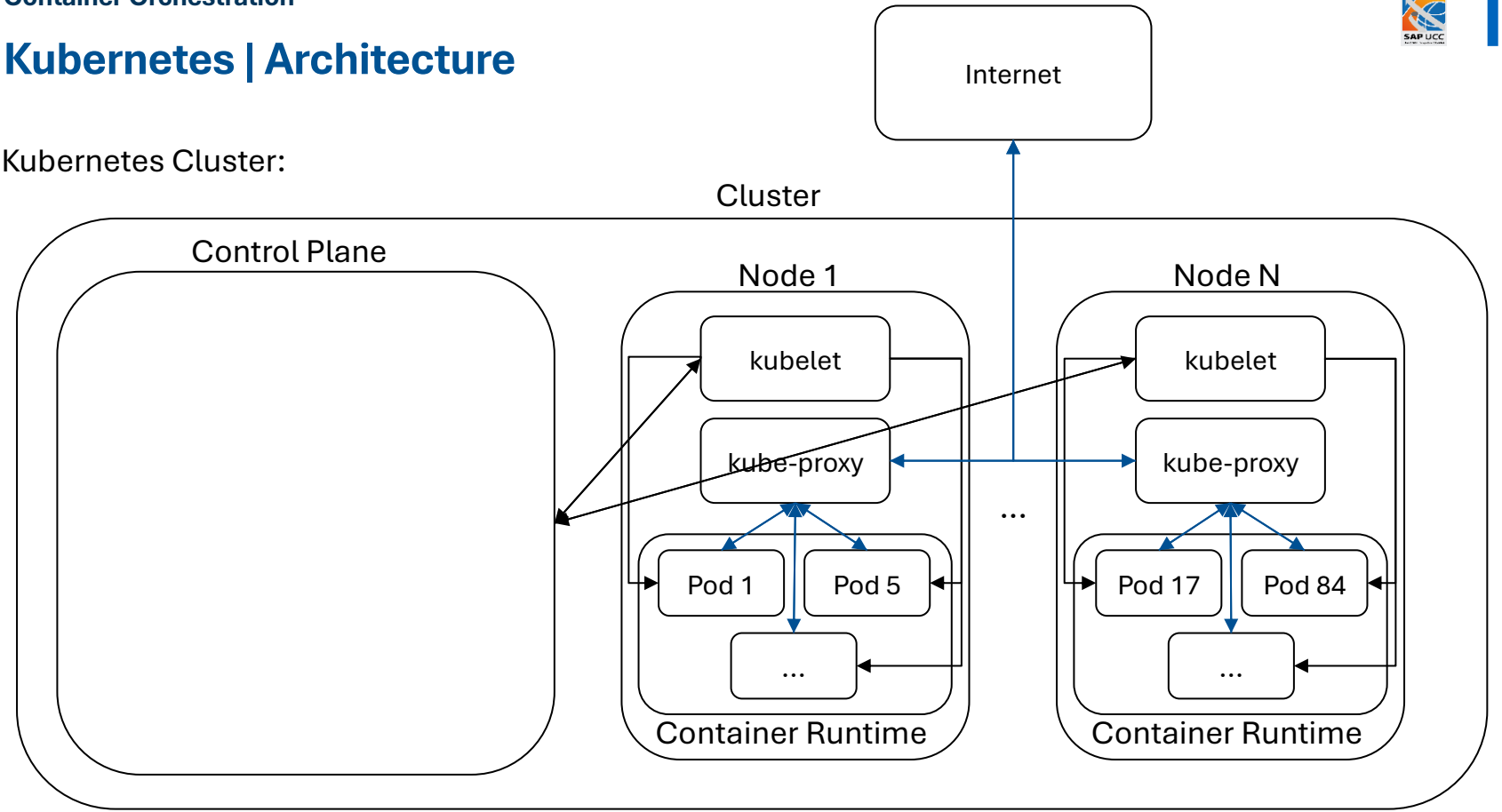

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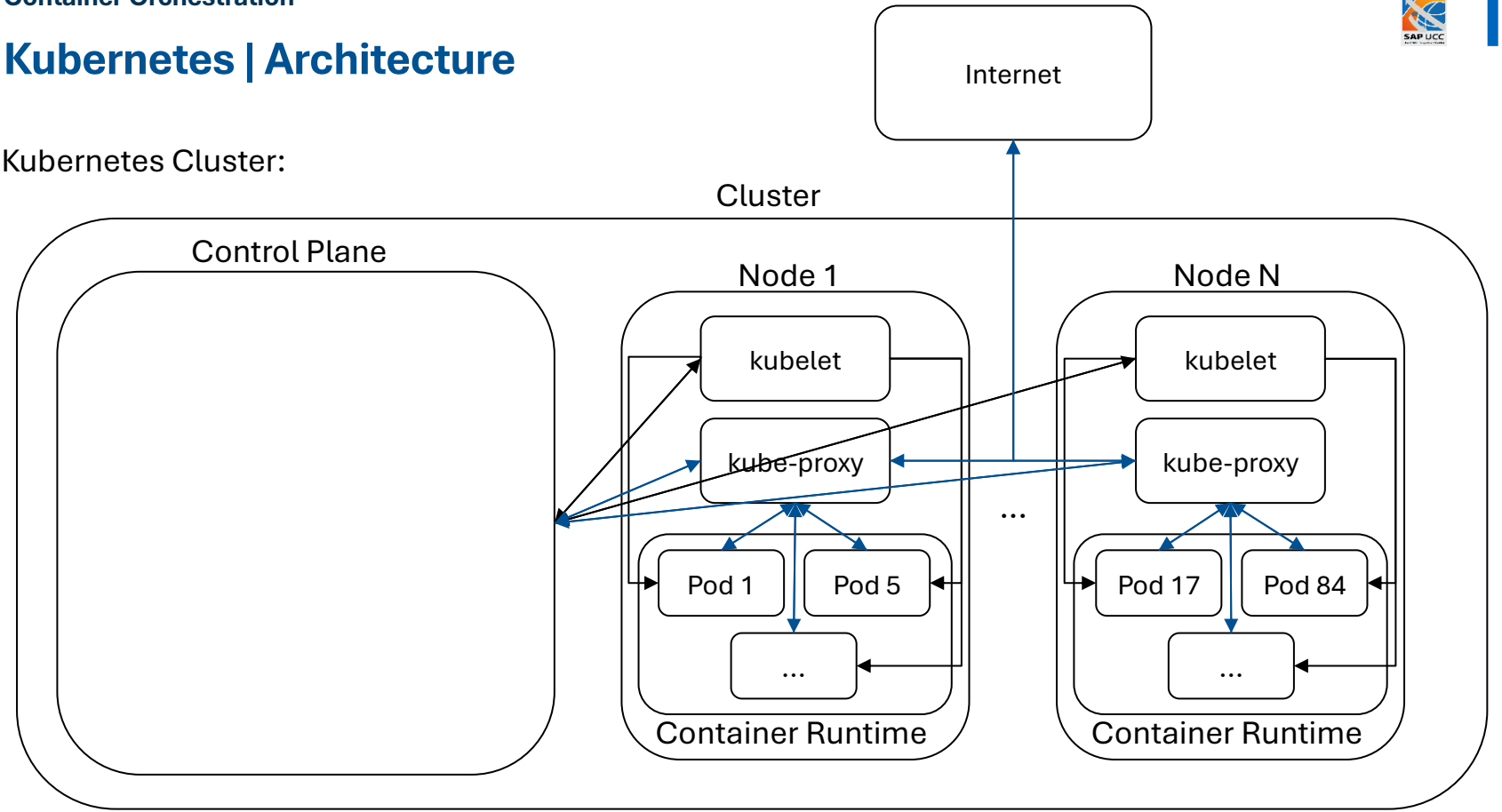

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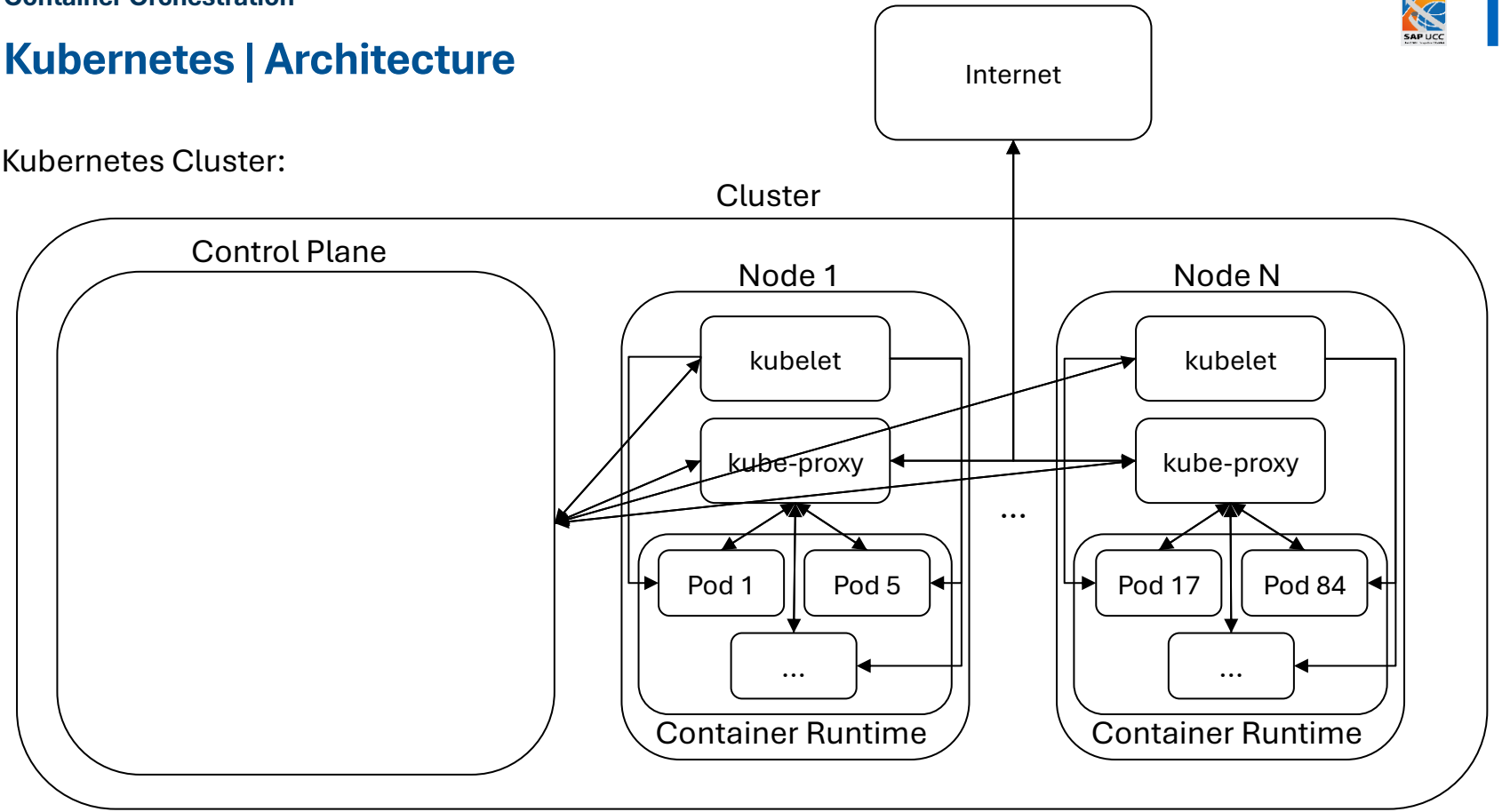

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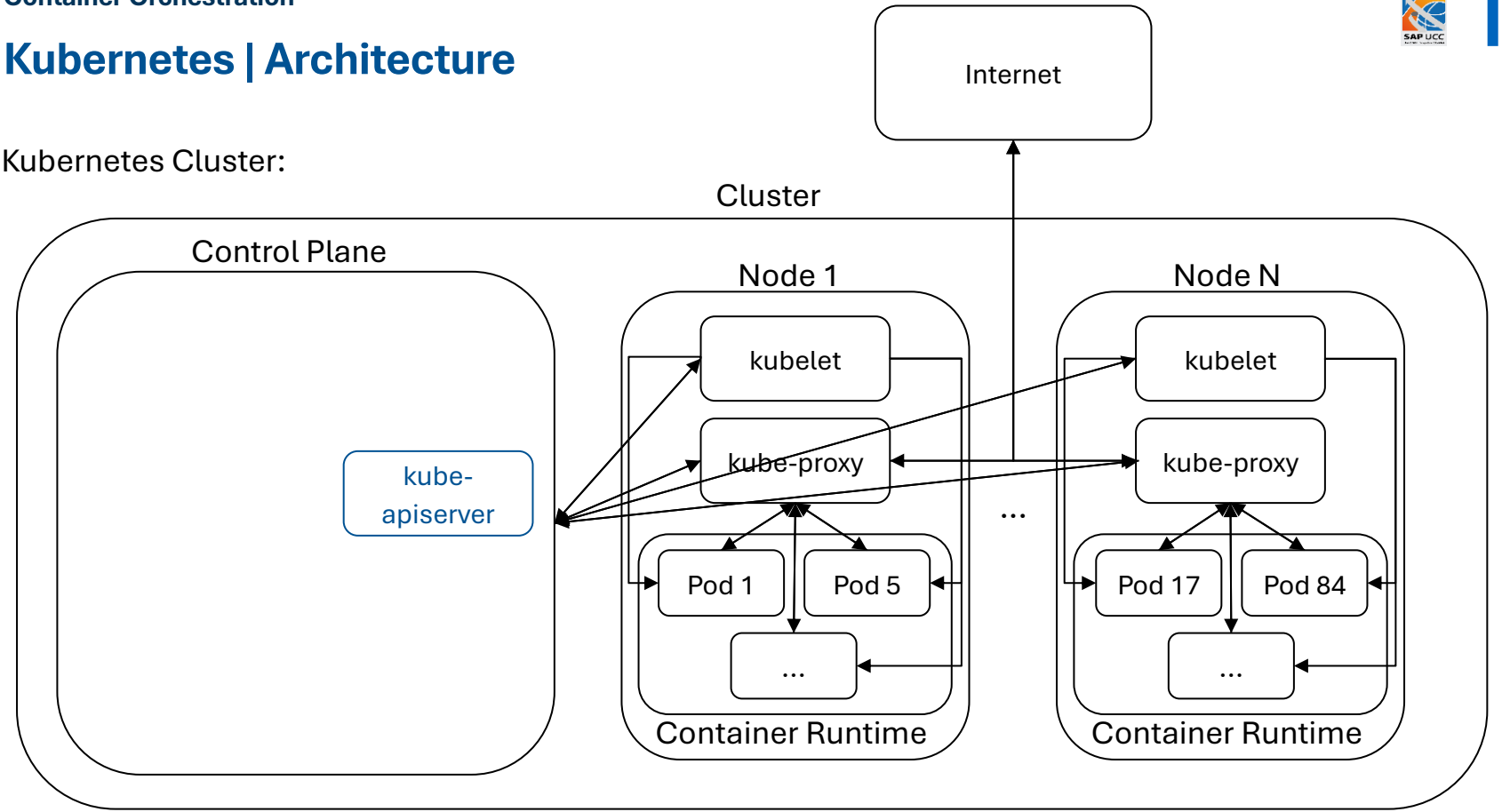


Based on: https://kubernetes.io/docs/concepts/architecture/

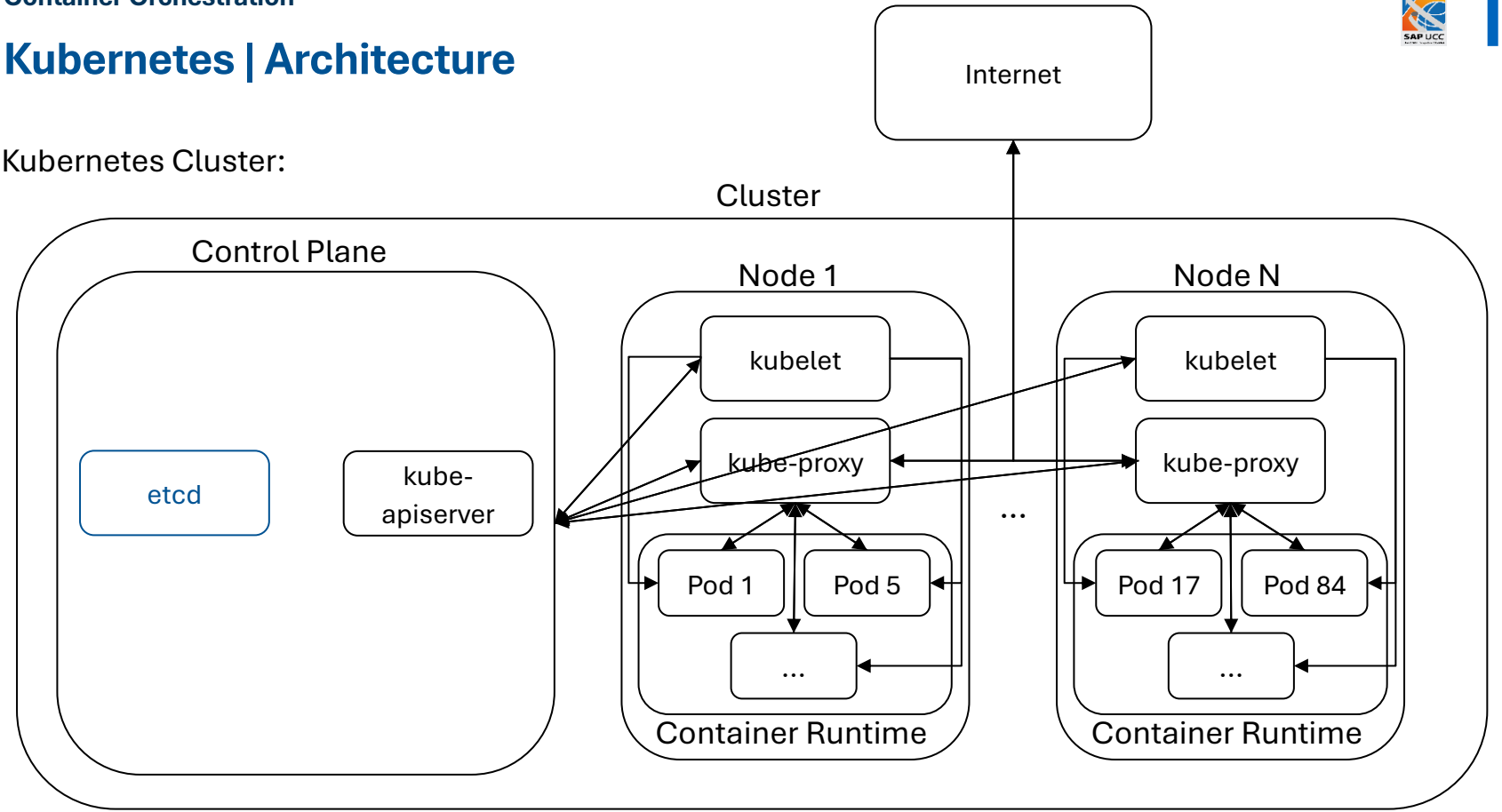
Kubernetes | Architecture

kube-apiserver:

- Exposes **Kubernetes API (REST)** to other components and users
- Frontend for control plane
- More instances can be deployed → horizontal scaling

Kubernetes | Architecture

Kubernetes Cluster:



Based on: https://kubernetes.io/docs/concepts/architecture/

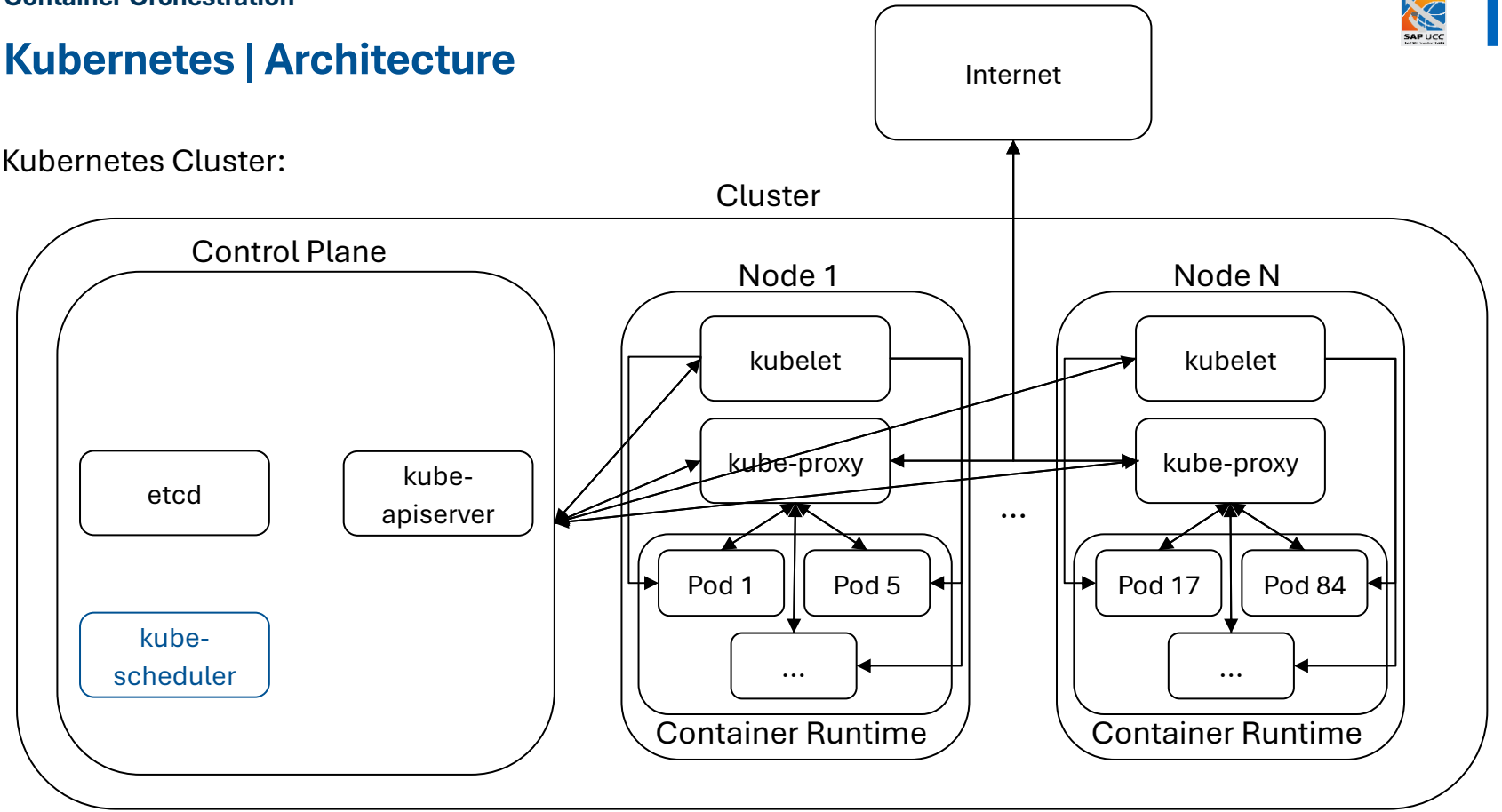
Kubernetes | Architecture

etcd:

- Distributed key-value store
- Stores **all cluster data**
- Made for high availability and consistency

Kubernetes | Architecture

Kubernetes Cluster:



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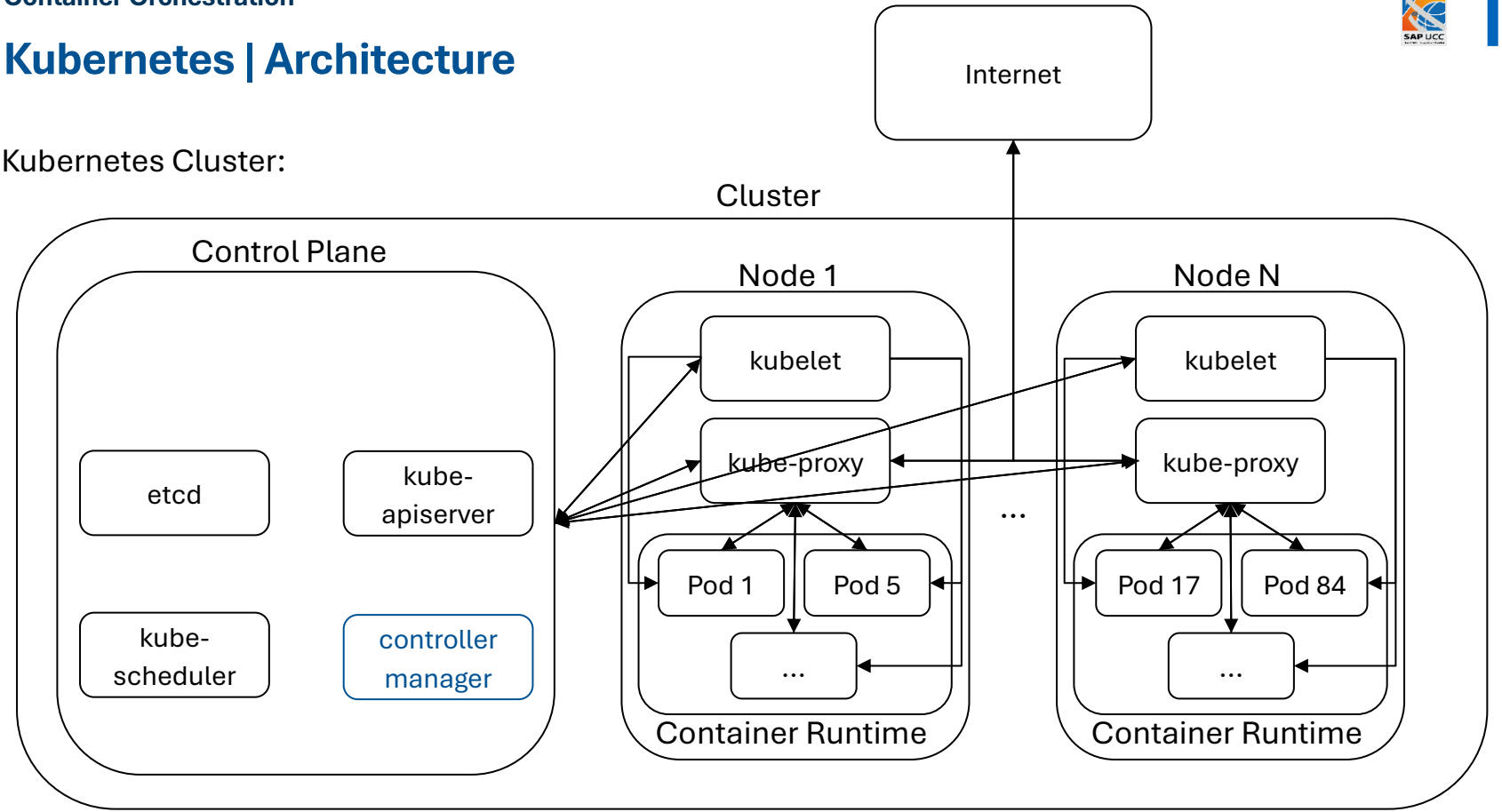
Kubernetes | Architecture

kube-scheduler:

- Watches for new Pods without designated node → **assigns node to run on**
- Node selection factors:
  - Resource requirements
  - Hardware and software policies
  - Inter workload interference

Kubernetes | Architecture

Kubernetes Cluster:



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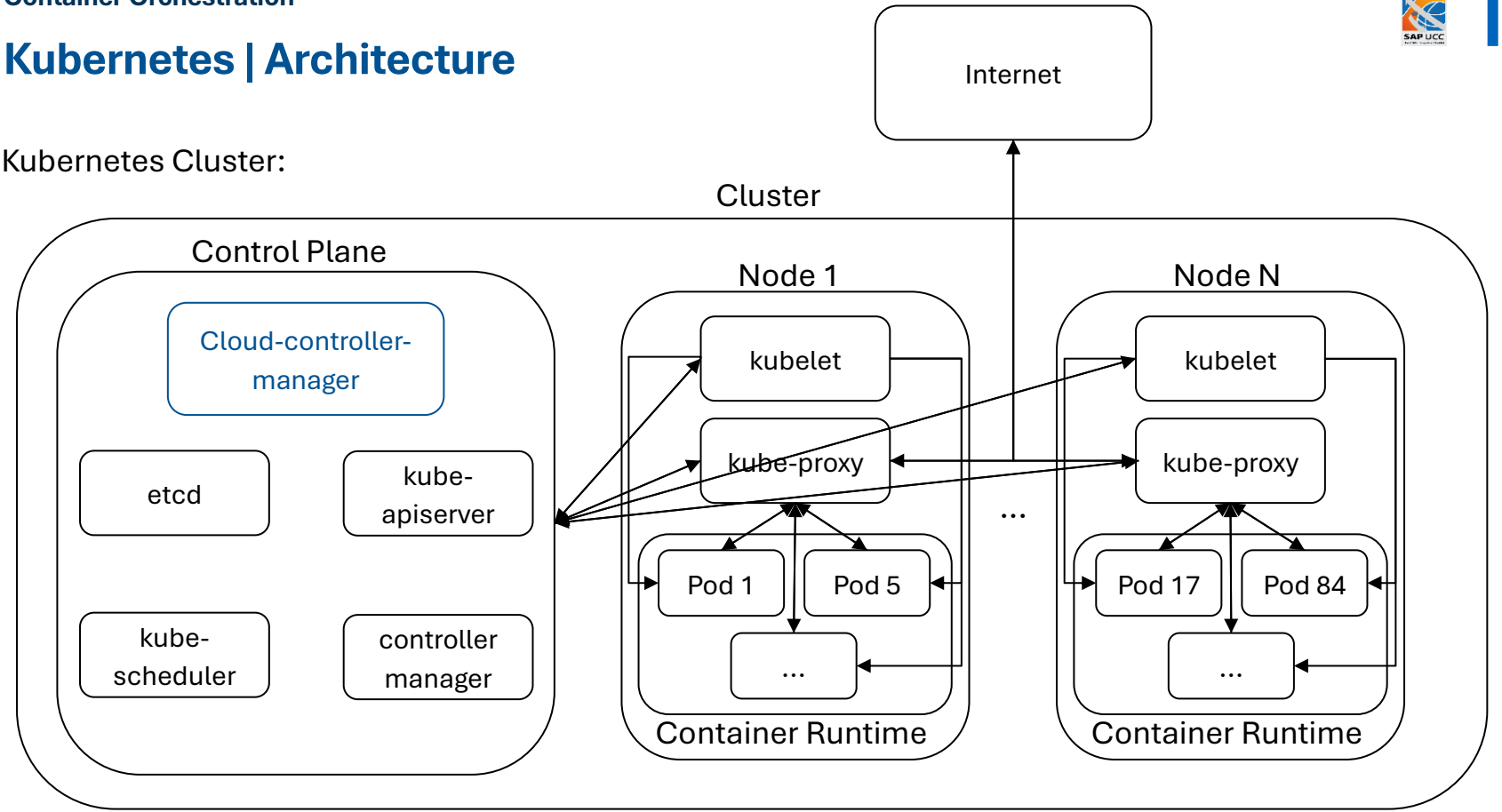
Kubernetes | Architecture

(kube-)controller-manager:

- Runs controller processes
- Controllers are compiled into one binary but are logically separate
- Example controllers:
  - Node controller: monitors if nodes are up and responds in case of downtime
  - Job controller: watches for Job objects (one-off tasks) → creates Pods to run those tasks
  - …

Kubernetes | Architecture

Kubernetes Cluster:



Based on: https://kubernetes.io/docs/concepts/architecture/

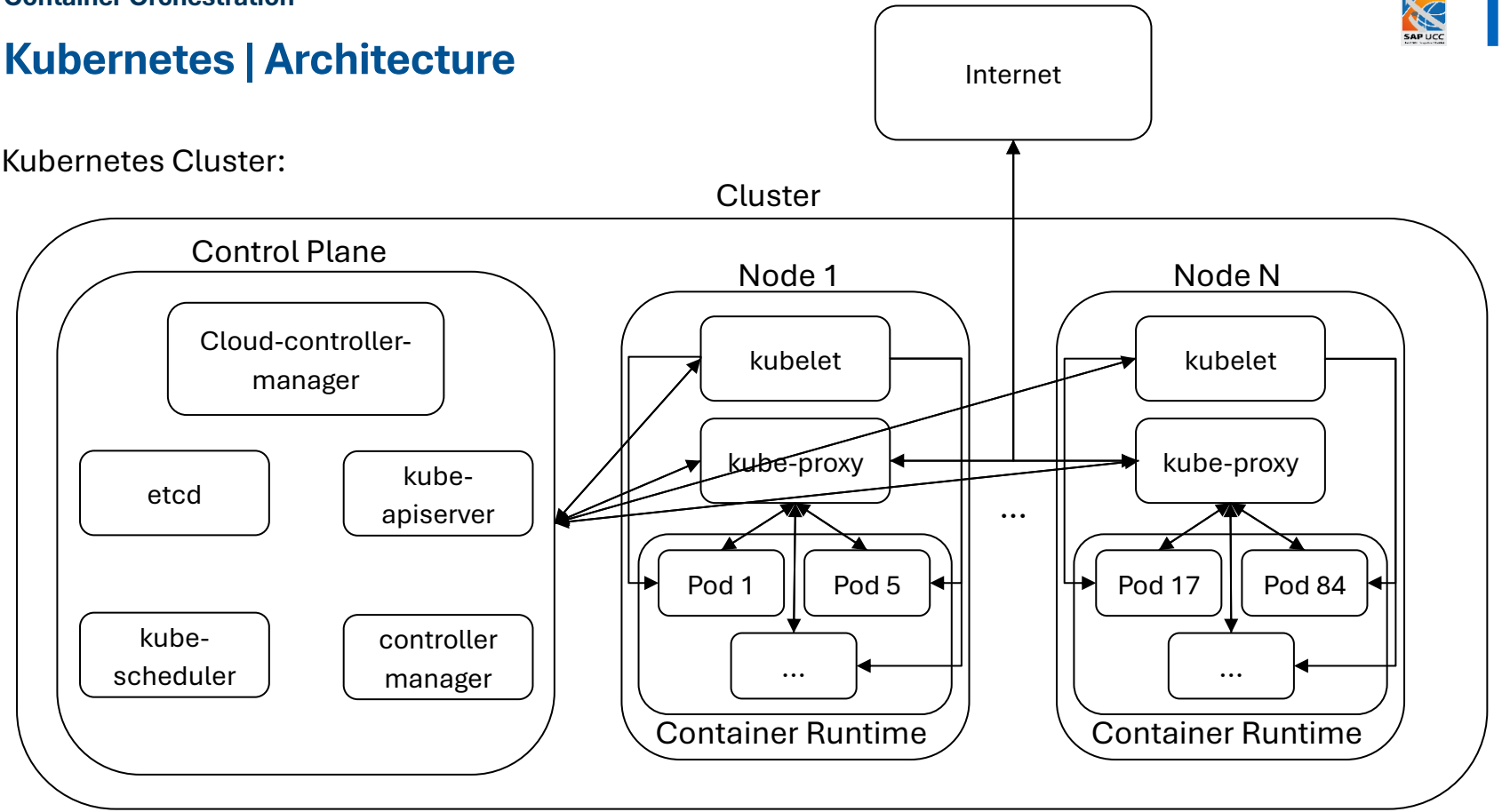
Kubernetes | Architecture

cloud-controller-manager:

- Allows linking a cluster into a cloud provider‘s API
- **Separates** components that interact with the **cloud** from objects that interact with the **local cluster**
- Runs cloud provider specific controllers, e.g.:
  - Route controller: sets up routes in cloud infrastructure
  - Service controller: creates, updates and deletes cloud provider load balancers
  - …

Kubernetes | Architecture

Kubernetes Cluster:



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Kubernetes | Architecture

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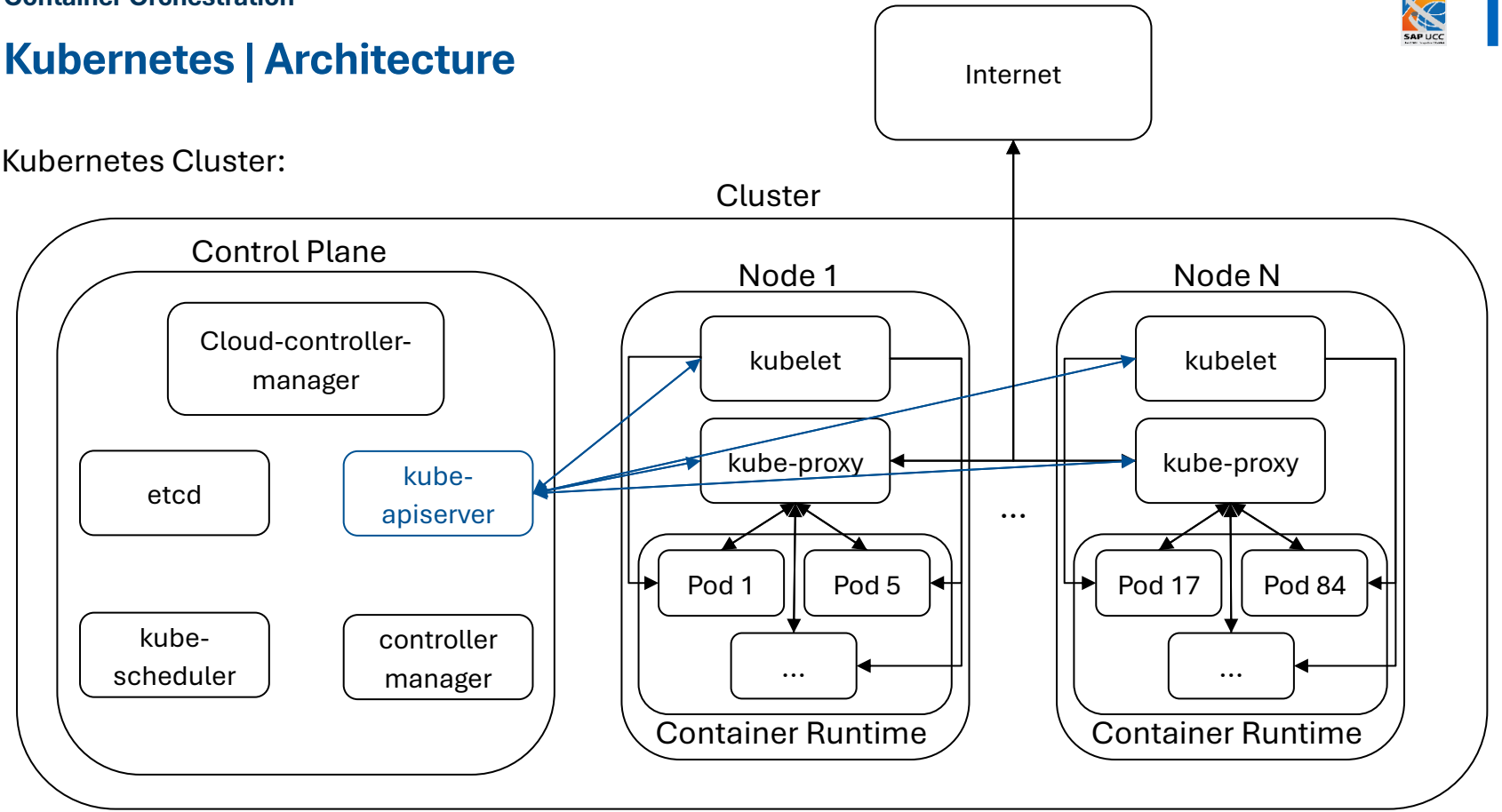

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Kubernetes | Architecture

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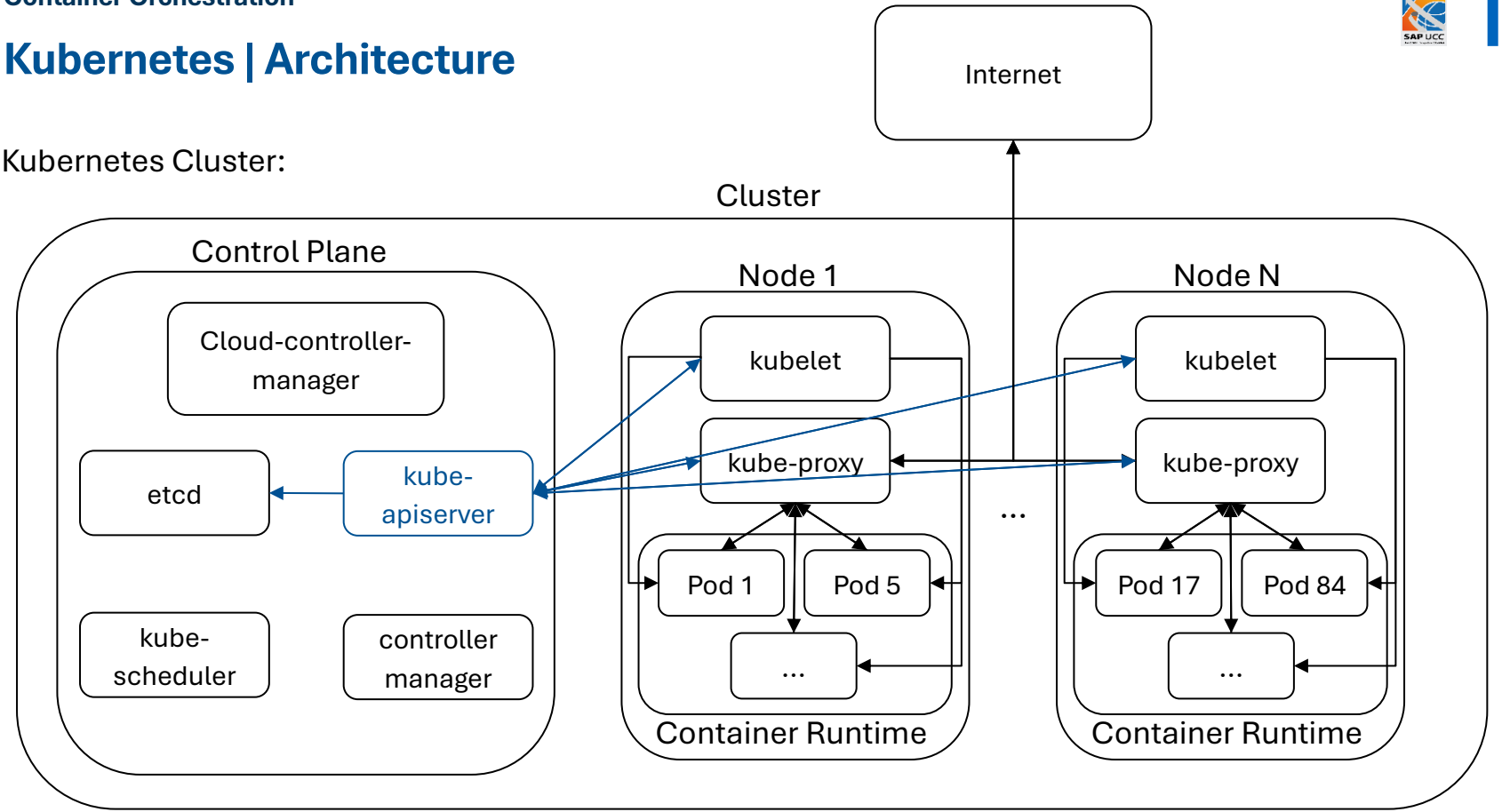

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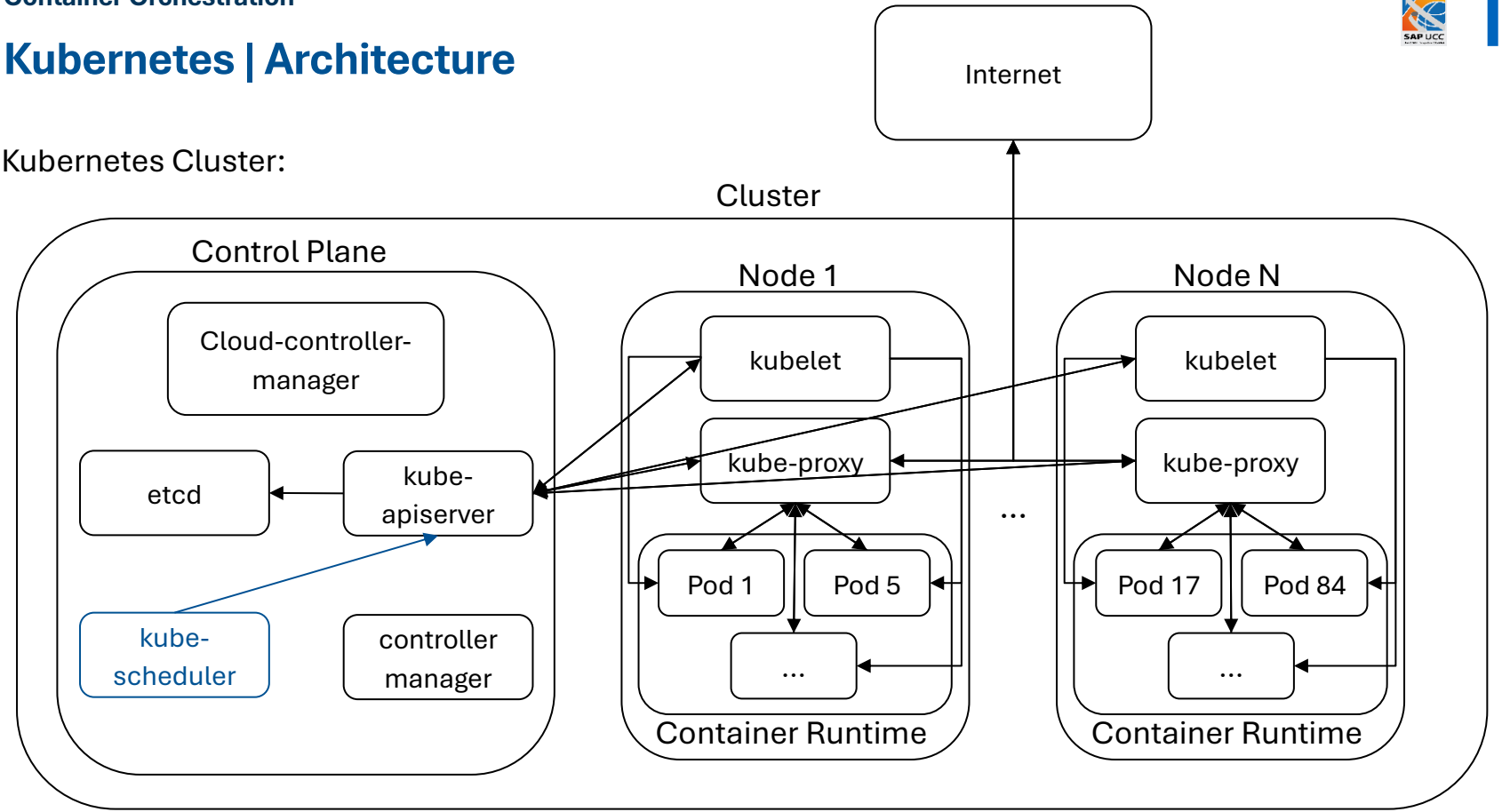

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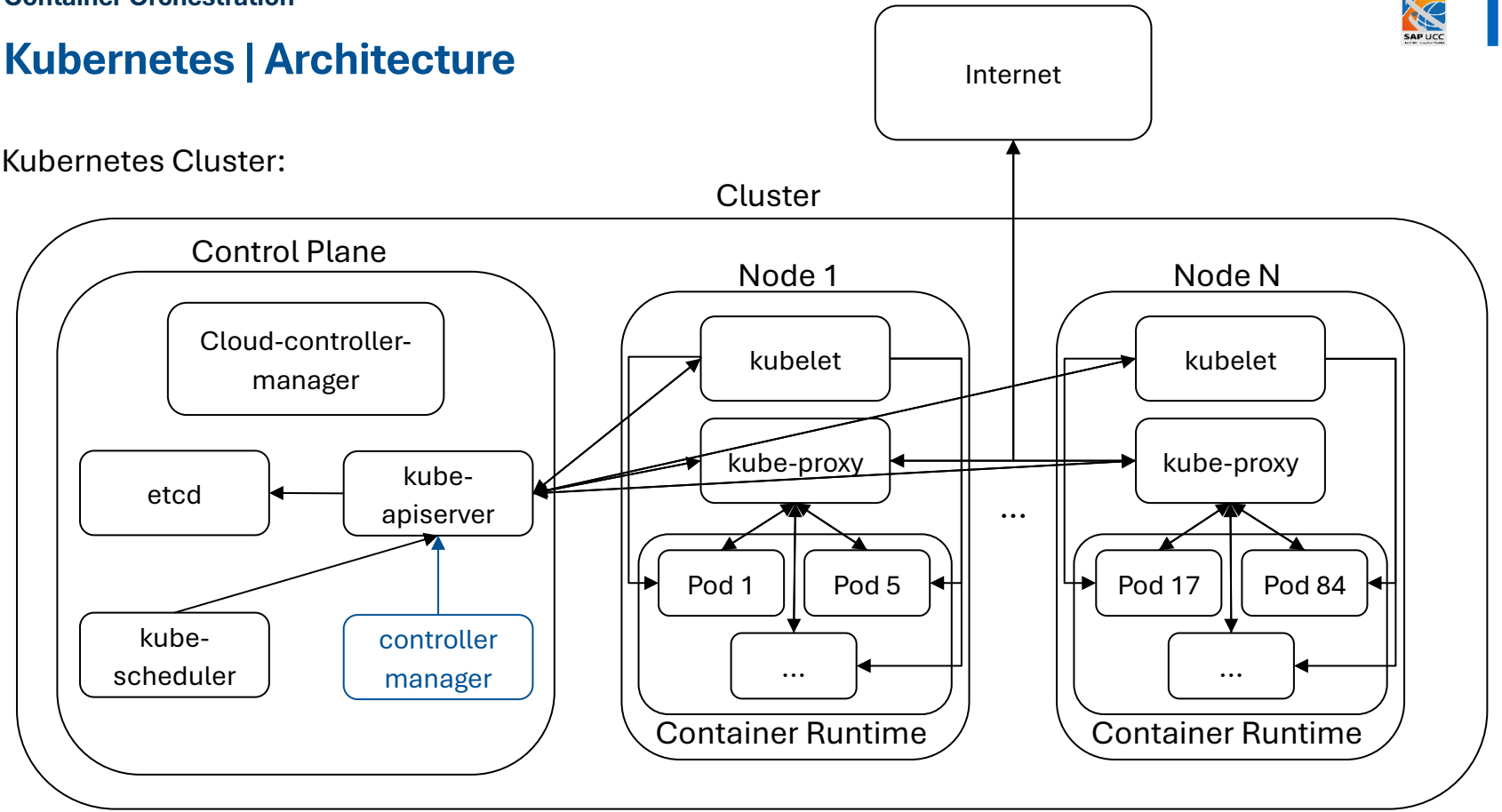

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Kubernetes | Architecture

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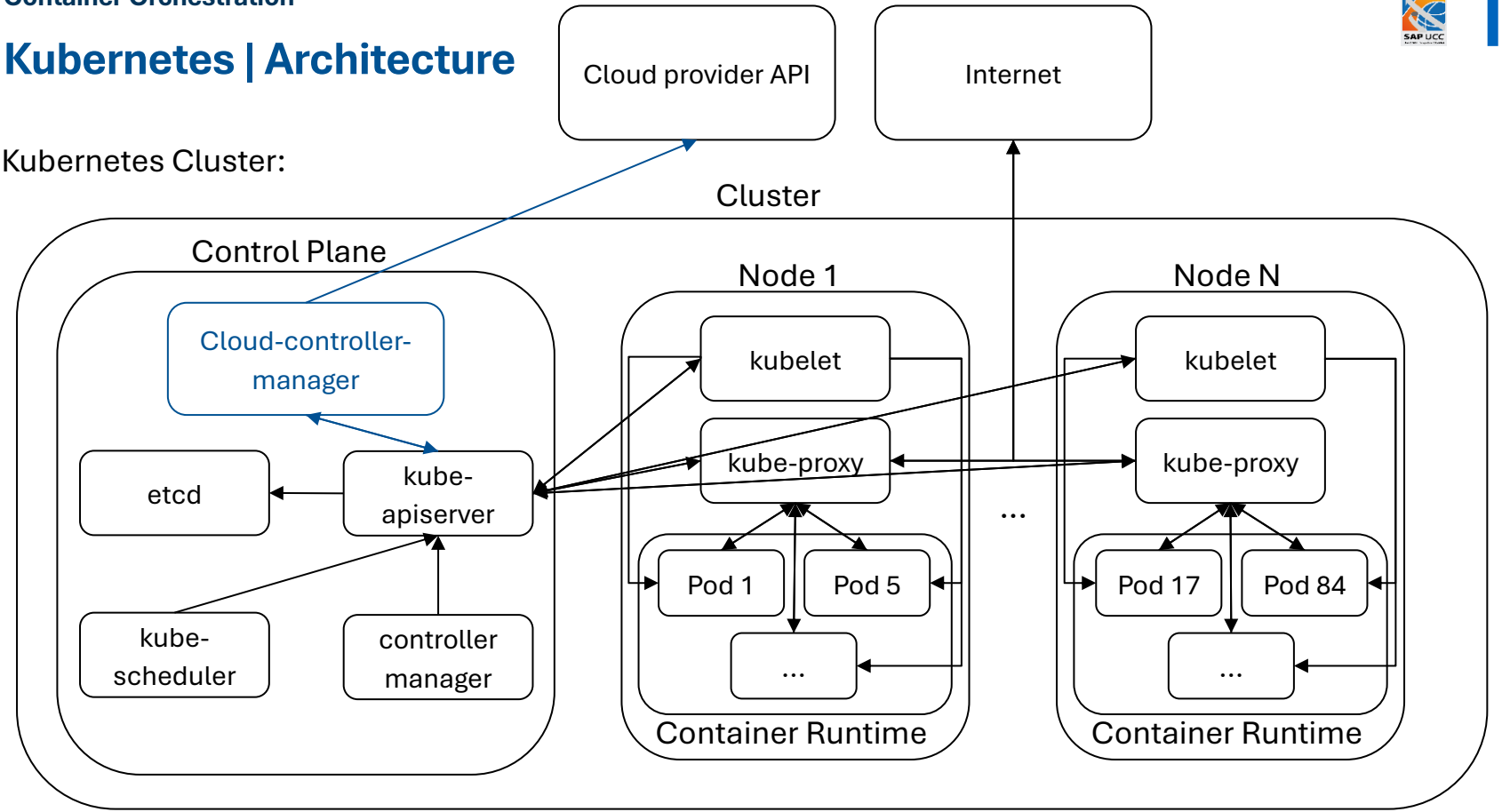

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Kubernetes | Architecture

Summary:

- Control plane components:
  - kube-apiserver → exposes Kubernetes API
  - etcd → stores cluster data
  - kube-scheduler → assigns Pods to nodes
  - kube-controller-manager → runs controller processes
  - cloud-controller-manager → runs controller processes related to cloud providers

Kubernetes | Architecture

Summary:

- Node components:
  - kubelet → communicates with Kubernetes-API and ensures given Pods are running
  - kube-proxy → facilitates network communication with pods
  - Container runtime → executes containers
  - Pods →smallest deployable unit, encapsulate containers

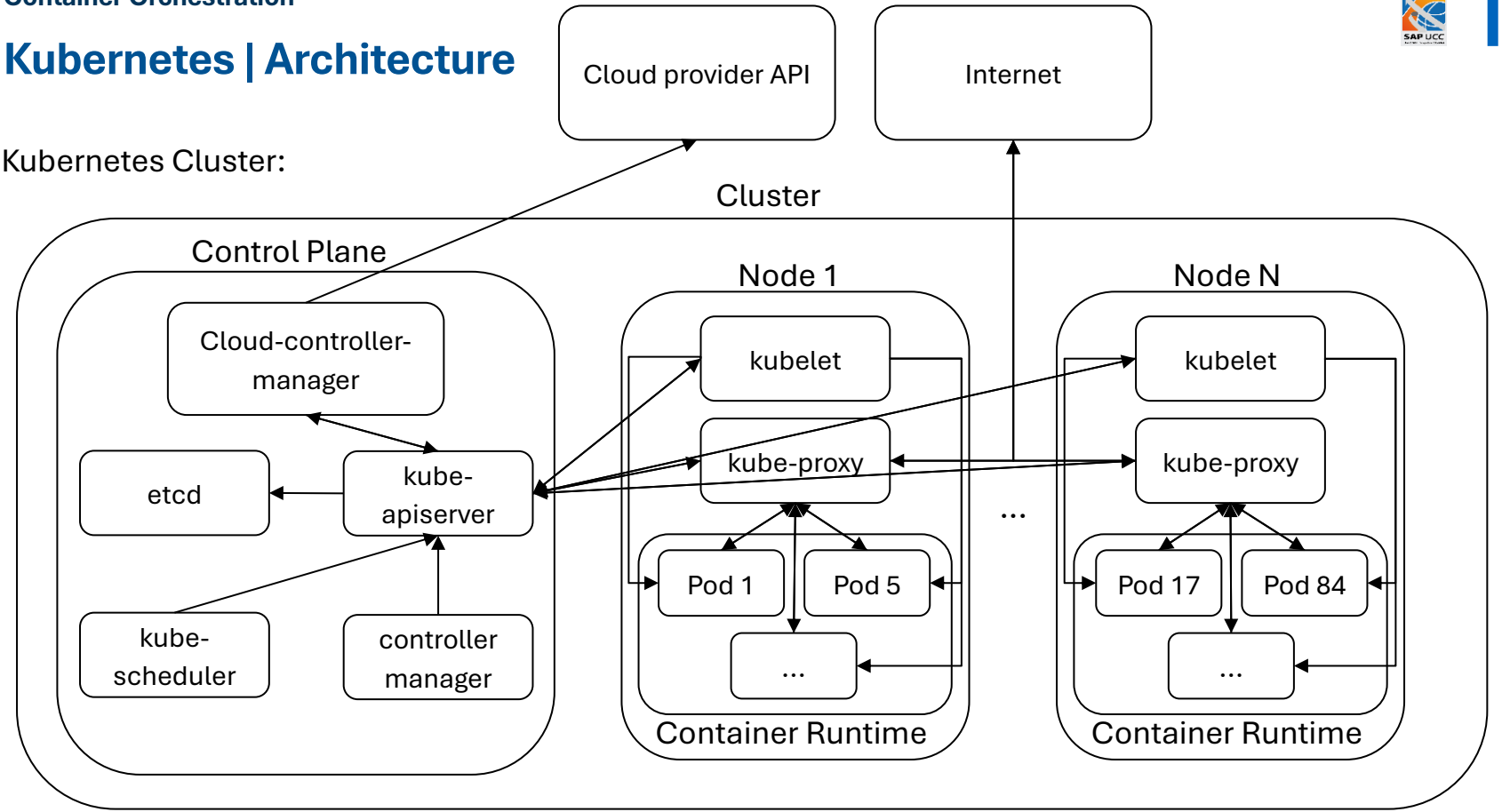
Kubernetes | Architecture

Cluster components:

- Can be deployed in containers (unless it runs containers, i.e., kubelet)
- Kubernetes can manage components running in containers itself
- Clusters can be extended by addons (e.g., DNS, Dashboard, ...)

Kubernetes | Architecture

Kubernetes Cluster:



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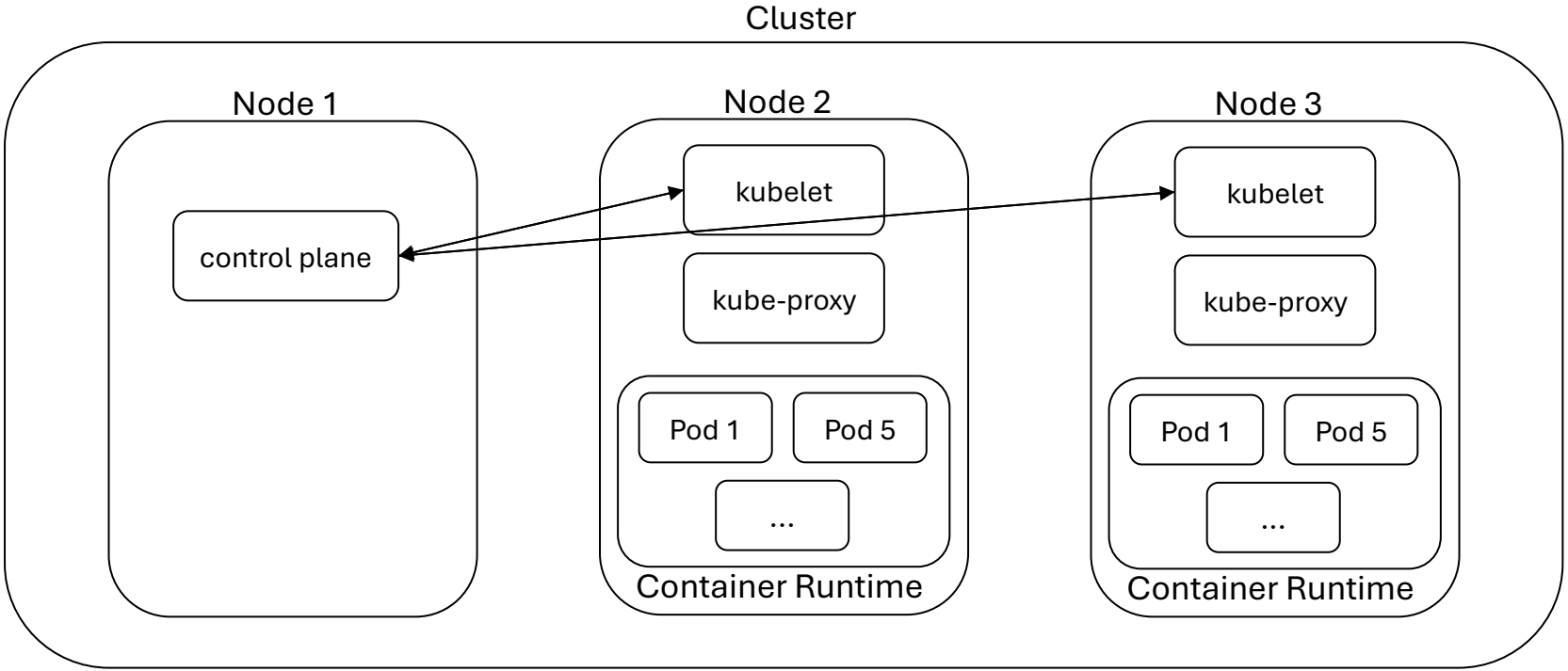
Kubernetes | Architecture

Example Setups:



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Kubernetes | Architecture

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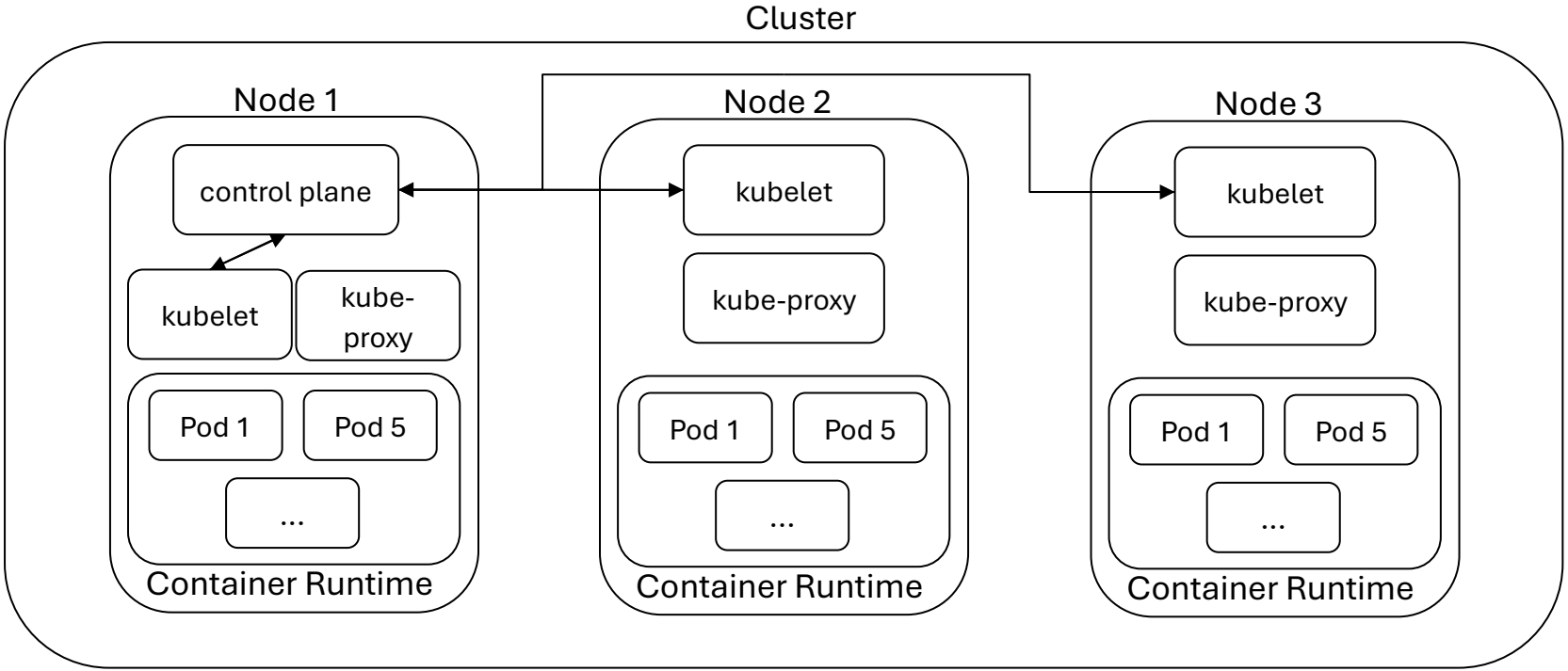

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Kubernetes | Architecture

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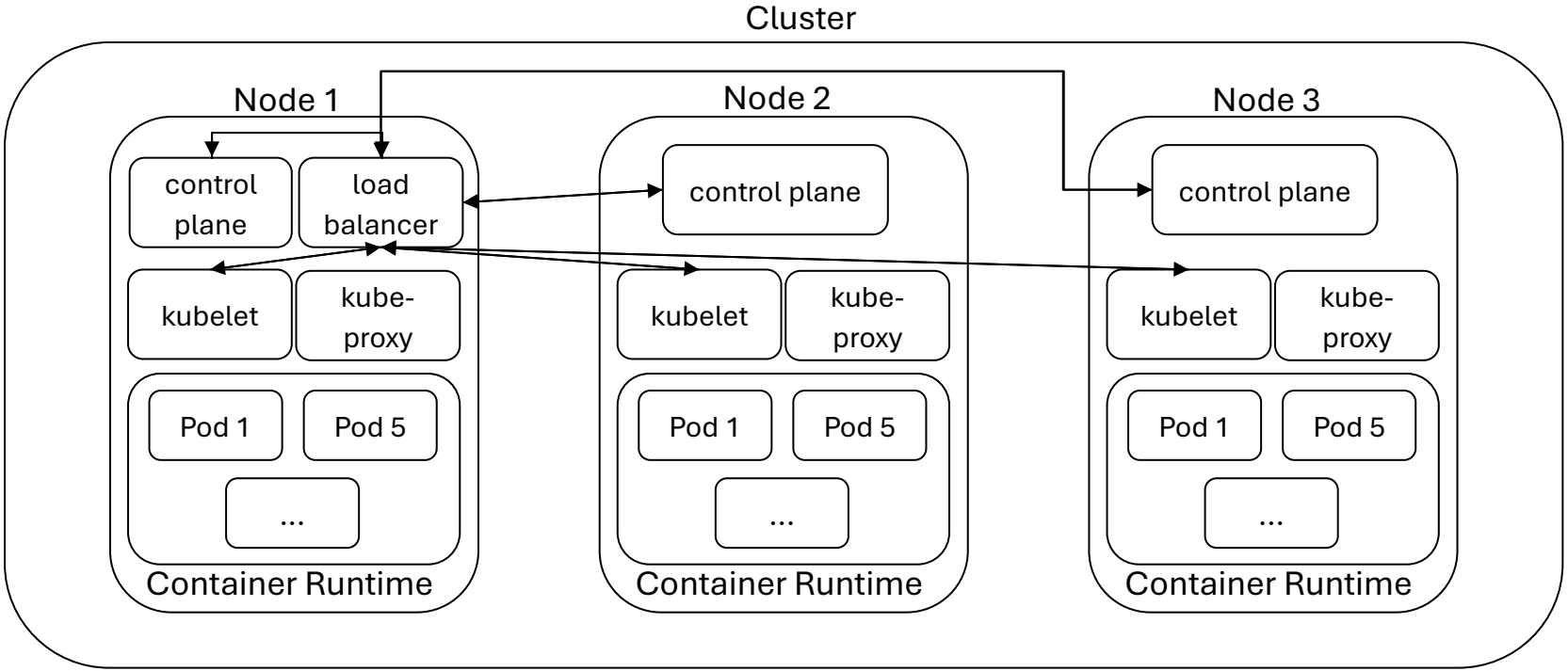

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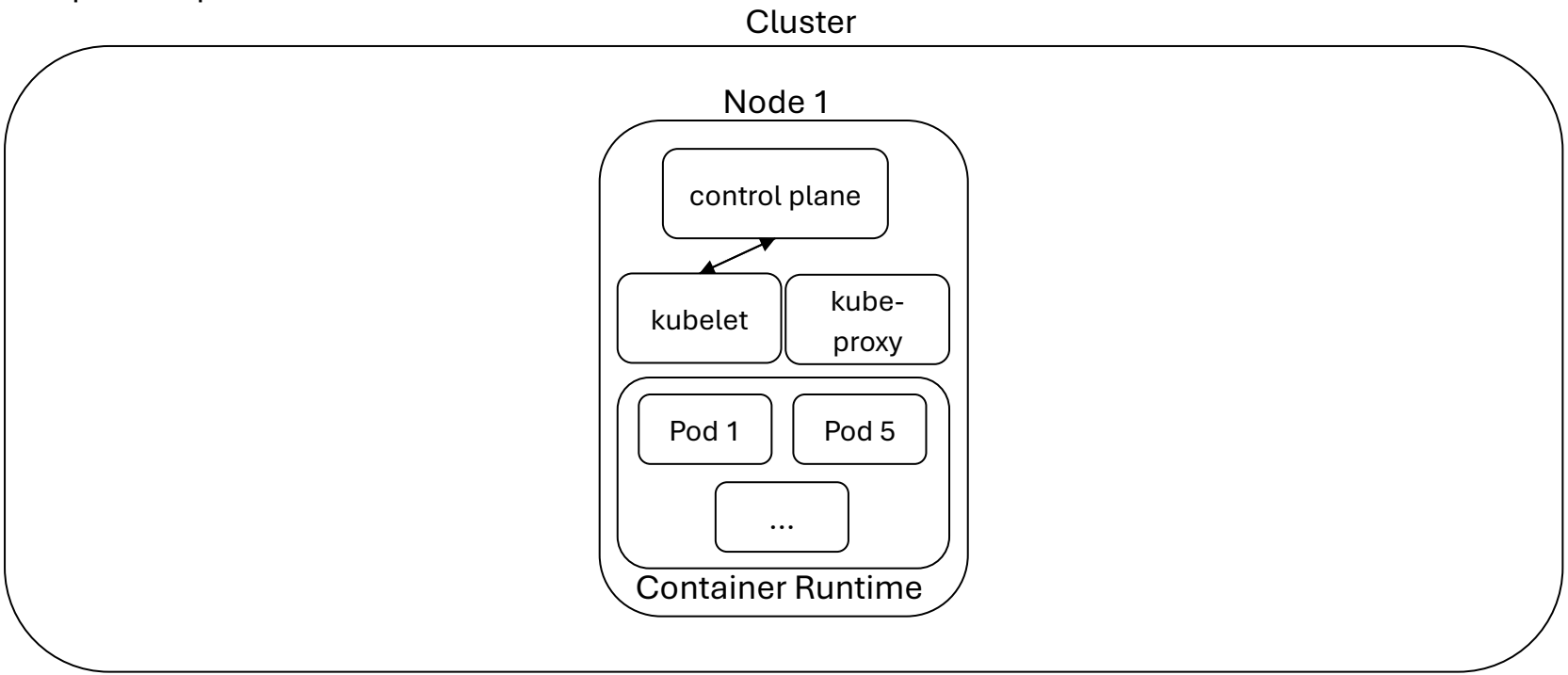
Kubernetes | Architecture

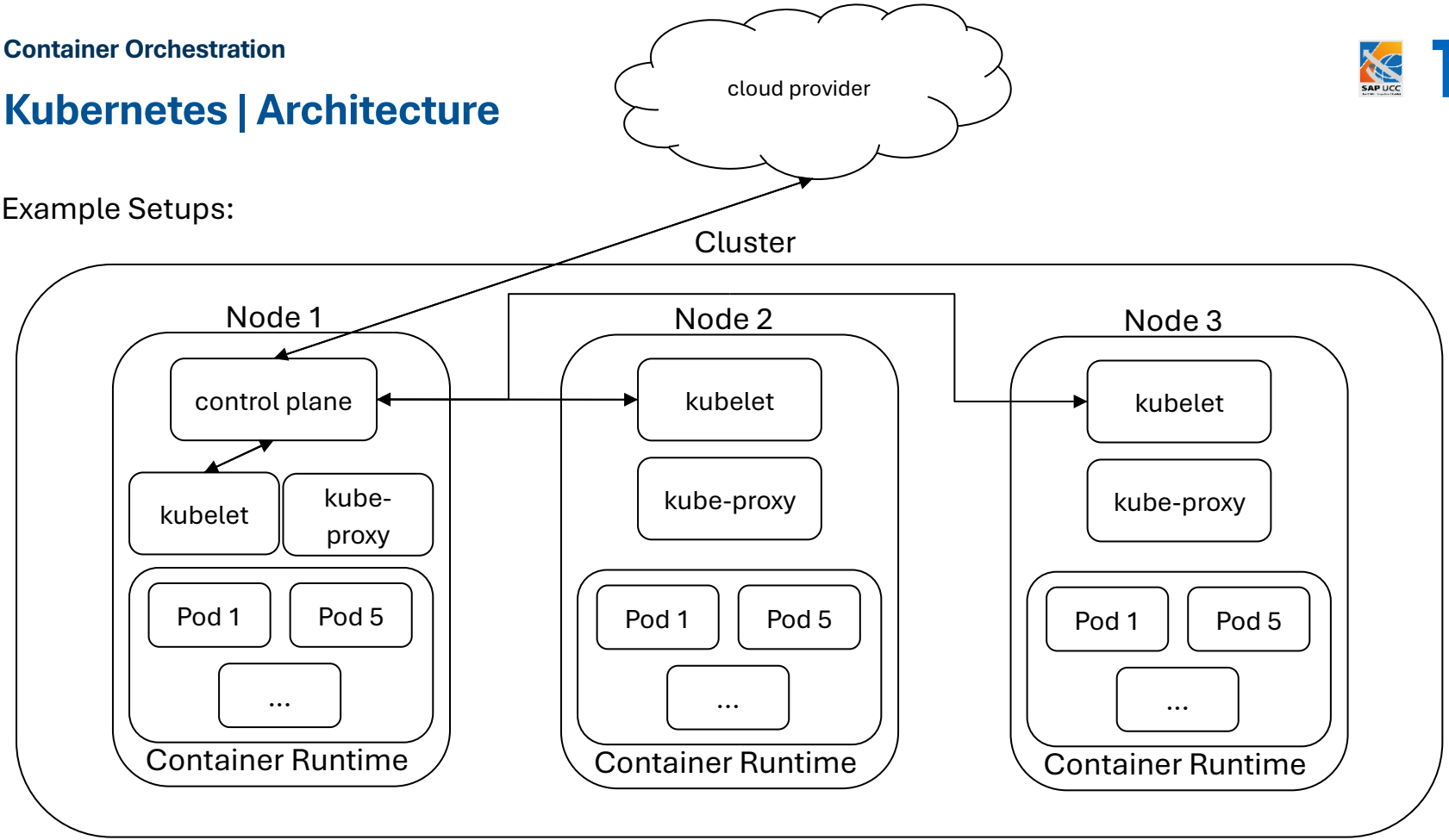
Example Setups:



Based on: https://kubernetes.io/docs/concepts/architecture/

Kubernetes | Architecture

Example Setups:



Based on: https://kubernetes.io/docs/concepts/architecture/

Kubernetes | Architecture

Cluster Capabilities:

- Kubernetes is designed to manage up to:
  - 5000 nodes
  - 110 Pods per node
  - 150,000 total Pods
  - 300,000 total containers

Kubernetes | Architecture

Kubernetes Partners:

IBM Cloud

Sources: see references

Kubernetes | Features

We have learned:

- What Kubernetes is
- Kubernetes main components
- Who uses Kubernetes

What we have not learned yet:

Kubernetes | Features

We have learned:

- What Kubernetes is
- Kubernetes main components
- Who uses Kubernetes

What we have not learned yet:

- How to use Kubernetes

Kubernetes | Features

minikube:

- Tool for setting up a **local** (single-node) Kubernetes cluster
- User-friendly (single command setup)
- Supports the latest Kubernetes version
- Open source (https://github.com/kubernetes/minikube/)
- Deploy cluster on VM, container, or bare metal

Sources: https://github.com/kubernetes/minikube/blob/master/images/logo/logo.svg

Kubernetes | Features

minikube:

We can start a local Kubernetes cluster using:

- “minikube start”

Sources: https://github.com/kubernetes/minikube/blob/master/images/logo/logo.svg

Kubernetes | Features

minikube:

We can start a local Kubernetes cluster using:

- “minikube start”

We can receive information about the cluster using:

- “minikube status”

Sources: https://github.com/kubernetes/minikube/blob/master/images/logo/logo.svg

Kubernetes | Features

kubeadm:

- Tool provided by Kubernetes
- Provides functionality to set up clusters
- Follows “best practice”
- Also intended as building block for higher level tools

Sources: https://github.com/kubernetes/website/blob/main/static/images/kubeadm-stacked-color.png

Kubernetes | Features

Kubernetes:

We can interact with the cluster using:

- “kubectl”

Command line interface:

- Enables communication with the control plane
- Utilizes the Kubernetes API

Source: https://github.com/kubernetes/kubernetes/blob/master/logo/logo.svg

Kubernetes | Features

Kubernetes:

We can see running Pods using:

- “kubectl get pods”
- “kubectl get pods -A” (Pods in all namespaces)

Source: https://github.com/kubernetes/kubernetes/blob/master/logo/logo.svg

Kubernetes | Features

Kubernetes Namespaces:

- Mechanism to isolate groups of resources in a cluster
- Names are required to be unique in a namespace
- Intended for multi-user or multi-team environments
- Cannot be nested
- Resources can only be in one namespace

Source: https://github.com/kubernetes/kubernetes/blob/master/logo/logo.svg

Kubernetes | Features

Kubernetes:

We can create Pods using:

- “kubectl run <*name*> --image=<*path-to-image*>”

Source: https://github.com/kubernetes/kubernetes/blob/master/logo/logo.svg

Kubernetes | Features

Kubernetes:

We can see create Pods using:

- “kubectl run <*name*> --image=<*path-to-image*>”

Alternatively, resources may be created using:

- “kubectl apply –f <*path-to-config-file*>“
- Creates objects from given configuration file (.yaml)

Source: https://github.com/kubernetes/kubernetes/blob/master/logo/logo.svg

Kubernetes | Features

Kubernetes:

Pod configuration file:

```
apiVersion: v1
kind: Pod
metadata:
  name: example-pod-2
spec:
  containers:
  - name: example-pod-2
    image: nginx:latest
    ports:
    - containerPort: 80
```

Source: https://github.com/kubernetes/kubernetes/blob/master/logo/logo.svg

Kubernetes | Features

Kubernetes:

We can find out information about objects (e.g., Pods) using:

- “kubectl describe <*resource-name*> <*object-name*>”

Source: https://github.com/kubernetes/kubernetes/blob/master/logo/logo.svg

Kubernetes | Features

Kubernetes:

To monitor and maintain our cluster, we can:

Retrieve logs of a Pod using:

- “kubectl logs <*object-name*>”

Execute commands in Pods using:

- “kubectl exec <*object-name*> -- <*command*>”

Source: https://github.com/kubernetes/kubernetes/blob/master/logo/logo.svg

Kubernetes | Features

Kubernetes:

To monitor and maintain our cluster, we can:

Forward system ports to a Pod using:

- “kubectl port-forward <*object-name*> <*port:port*>”

Source: https://github.com/kubernetes/kubernetes/blob/master/logo/logo.svg

Kubernetes | Features

Kubernetes:

We can remove resources using:

- “kubectl delete <*resource-name*> <*object-name*>”
- “kubectl delete –f <*path-to-config-file*>“

Source: https://github.com/kubernetes/kubernetes/blob/master/logo/logo.svg

Kubernetes | Features

Labels and Annotations:

Objects can be labeled and annotated:

- Labels:
  - Used to identify and group objects
  - Can be used as selector when listing objects (--selector="...")
- Annotations: key-value pairs, store information

Source: https://github.com/kubernetes/kubernetes/blob/master/logo/logo.svg

Kubernetes | Features

Labels and Annotations:

Labeled Pod configuration file:

```
apiVersion: v1
kind: Pod
metadata:
  name: example-pod-2
  labels:
    env: "prod"
spec:
  containers:
  - name: example-pod-2
    image: nginx:latest
    ports:
    - containerPort: 80
```

Source: https://github.com/kubernetes/kubernetes/blob/master/logo/logo.svg

Kubernetes | Features

Resource constraints:

Kubernetes can set bounds on the resources assigned to Pods:

- Requests: lower bound on resources
- Limit: upper bound on resources

Kubernetes can restrict *memory* and *cpu* usage

Source: https://github.com/kubernetes/kubernetes/blob/master/logo/logo.svg

Kubernetes | Features

Resource constraints:

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: example-pod-2-constrained
  labels:
    env: "prod"
spec:
  containers:
  - name: example-pod-2-constrained
    image: nginx:latest
    resources:
      requests:
        memory: "64Mi"
        cpu: "250m"
      limits:
        memory: "128Mi"
        cpu: "500m"
    ports:
    - containerPort: 80
```

Source: https://github.com/kubernetes/kubernetes/blob/master/logo/logo.svg

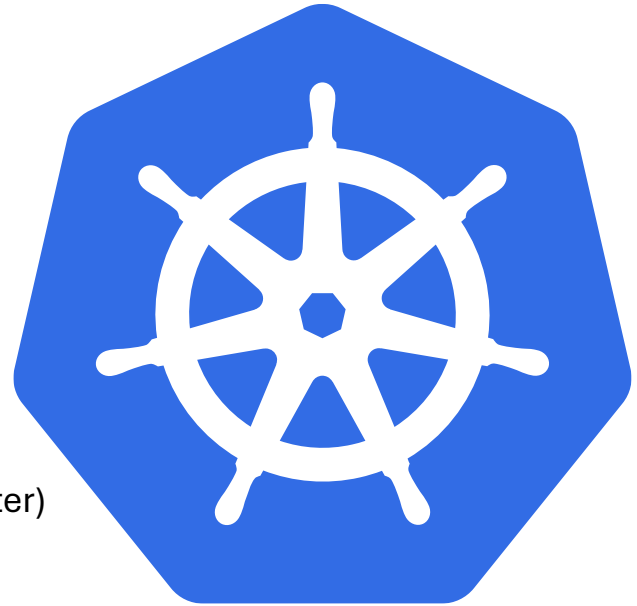
Kubernetes | Features

Services & Service Discovery:

- Kubernetes is very dynamic → IP addresses of Pods may change
- Pods may be running on multiple nodes
- Requires decoupling of network addresses for Pods from nodes

→ Services expose network applications (inside and outside the cluster)

Cluster-aware DNS servers create DNS records for new services



Source: https://github.com/kubernetes/kubernetes/blob/master/logo/logo.svg

Kubernetes | Features

Services & Service Discovery:

We can create services using:

- “kubectl expose <*resource-name*> <*object-name*>”
- “kubectl apply –f <*path-to-config-file*>“

Source: https://github.com/kubernetes/kubernetes/blob/master/logo/logo.svg

Kubernetes | Features

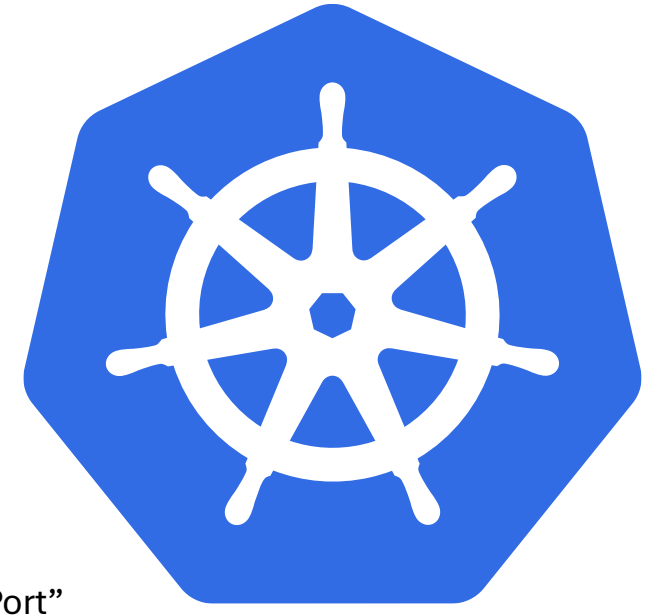
Services & Service Discovery:

We can create services using:

- “kubectl expose <*resource-name*> <*object-name*>”
- “kubectl apply –f <*path-to-config-file*>“

To expose the service outside the cluster:

- “kubectl expose <*resource-name*> <*object-name*> --type=NodePort”



Source: https://github.com/kubernetes/kubernetes/blob/master/logo/logo.svg

Kubernetes | Features

ReplicaSets:

Some applications require multiple concurrent instances, e.g., for:

- Redundancy, Scalability, or Sharding

ReplicaSets manage a set of replicas of a single Pod:

- Ensure a given number of instances are running
- Can be configured to scale automatically

Source: https://github.com/kubernetes/kubernetes/blob/master/logo/logo.svg

Kubernetes | Features

ReplicaSets:

```
apiVersion: apps/v1
kind: ReplicaSet
metadata:
  name: example-replica
  labels:
    env: prod
spec:
  replicas: 3
  selector:
    matchLabels:
      env: prod
  template:
    metadata:
      labels:
        env: prod
    spec:
      containers:
      - name: example-pod-3
        image: nginx:latest
```

Source: https://github.com/kubernetes/kubernetes/blob/master/logo/logo.svg

Kubernetes | Features

ReplicaSets:

We can scale ReplicaSets using:

- “kubectl scale replicasets <*object-name*> --replicas=<*number*>”
- “kubectl apply –f <*path-to-config-file*>“

Source: https://github.com/kubernetes/kubernetes/blob/master/logo/logo.svg

Kubernetes | Features

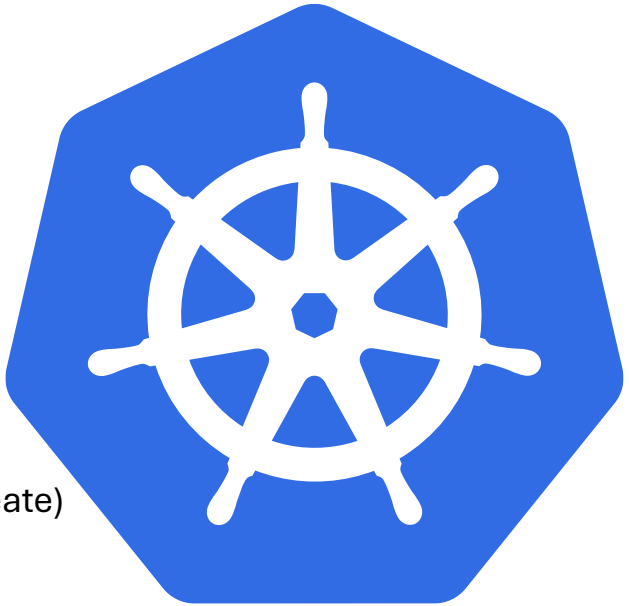
Deployments:

Pods and ReplicaSets are tied to a given container image:

- Need to be fully removed and re-created for updates

Deployments:

- Facilitate rollout process for new updates (Rolling update or Recreate)
- Manage ReplicaSets



Source: https://github.com/kubernetes/kubernetes/blob/master/logo/logo.svg

Kubernetes | Features

Deployments :

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: example-deployment
  labels:
    env: prod
spec:
  replicas: 3
  selector:
    matchLabels:
      env: prod
  template:
    metadata:
      labels:
        env: prod
    spec:
      containers:
      - name: example-pod-4
        image: nginx:latest
  strategy:
    type: RollingUpdate
    rollingUpdate:
      maxUnavailable: 1
```

Source: https://github.com/kubernetes/kubernetes/blob/master/logo/logo.svg

Kubernetes | Features

Deployments :

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: example-deployment
  labels:
    env: prod
spec:
  replicas: 3
  selector:
    matchLabels:
      env: prod
  template:
    metadata:
      annotations:
        kubernetes.io/change-cause: "Some change"
      labels:
        env: prod
    spec:
      containers:
      - name: example-pod-4
        image: nginx:1.28
  strategy:
    type: RollingUpdate
    rollingUpdate:
      maxUnavailable: 1
```

Source: https://github.com/kubernetes/kubernetes/blob/master/logo/logo.svg

Kubernetes | Features

Deployment:

We can manage the rollout of updates for Deployments using:

- “kubectl rollout status deployments <*object-name*>”
- “kubectl rollout pause deployments <*object-name*>”
- “kubectl rollout resume deployments <*object-name*>”
- “kubectl rollout history deployments <*object-name*>”

Source: https://github.com/kubernetes/kubernetes/blob/master/logo/logo.svg

Kubernetes | Features

DaemonSets:

Deployments and ReplicaSets:

- (Generally) agnostic about nodes they run on
- Not as useful for running cluster internal agents or daemons

DaemonSets:

- Run a Pod on each (or a subset) of the nodes in the cluster

Source: https://github.com/kubernetes/kubernetes/blob/master/logo/logo.svg

Kubernetes | Features

DaemonSets :

```
apiVersion: apps/v1
kind: DaemonSet
metadata:
  name: example-daemonset
  labels:
    env: prod
spec:
  selector:
    matchLabels:
      env: prod
  template:
    metadata:
      labels:
        env: prod
    spec:
      containers:
      - name: example-pod-5
        image: nginx:latest
```

Source: https://github.com/kubernetes/kubernetes/blob/master/logo/logo.svg

Kubernetes | Features

Jobs:

Deployments, ReplicaSets, and Daemonsets:

- Meant for active services and applications
- Restart containers on failure

Jobs → Represent Pods meant for one-off tasks

- Execute a Pod → do not restart

Source: https://github.com/kubernetes/kubernetes/blob/master/logo/logo.svg

Kubernetes | Features

Jobs :

```
apiVersion: batch/v1
kind: Job
metadata:
  name: example-job
spec:
  template:
    spec:
      containers:
      - name: example-job
        image: python:latest
        command:
        - python3
        - -c
        - |
          import os, sys
          print("Hello world")
      restartPolicy: Never
```

Source: https://github.com/kubernetes/kubernetes/blob/master/logo/logo.svg

Kubernetes

Summary:

- Hyperscalers require a very large number of containers (e.g., to run cloud services)
- Kubernetes can **deploy**, **scale**, and **manage** containerized applications
- Components:
  - Control plane: kube-apiserver, etcd, kube-scheduler, kube-controller-manager, cloud-controller-manager
  - Nodes: kubelet, kube-proxy, Container runtime, Pods

Kubernetes

Summary:

- Important Kubernetes objects:
  - Pods
  - Services
  - ReplicaSets
  - Deployments
  - DaemonSets
  - Jobs

Source: https://github.com/kubernetes/kubernetes/blob/master/logo/logo.svg

Sources

- https://docs.docker.com/get-started/docker-overview/
- https://github.com/opencontainers/image-spec/blob/main/spec.md
- https://opencontainers.org/
- https://docs.docker.com/reference/dockerfile
- https://www.man7.org/linux/man-pages/man7/namespaces.7.html
- https://linux.die.net/man/1/unshare
- https://www.man7.org/linux/man-pages/man7/cgroups.7.html
- https://docs.docker.com/get-started/docker-concepts/the-basics/what-is-a-registry/
- https://github.com/opencontainers/distribution-spec/blob/main/spec.md
- https://docs.docker.com/build/concepts/dockerfile/
- https://github.com/opencontainers/runtime-spec/blob/main/spec.md
- https://docs.docker.com/engine/daemon/alternative-runtimes/

Sources

- https://github.com/containers/podman
- https://podman.io/
- Hightower, K., Burns, B., Beda, J., & Demmig, T. (2018). *Kubernetes: Eine kompakte Einführung* (1. Auflage.). dpunkt.verlag.
- https://www.redhat.com/de/topics/cloud-computing/what-is-a-hyperscaler
- https://www.ibm.com/de-de/think/topics/hyperscale
- https://cloud.google.com/learn/what-is-a-cloud-service-provider?hl=en
- https://www.redhat.com/en/topics/cloud-computing/what-are-cloud-providers
- https://www.flaticon.com/free-icon/container_7115168?term=container&page=1&position=1&origin=tag&related_id=7115168
- https://github.com/orgs/networkx/repositories

Sources

- https://www.ibm.com/think/topics/kubernetes-history
- https://kubernetes.io/docs/setup/best-practices/cluster-large/
- https://queue.acm.org/detail.cfm?id=2898444
- https://www.cncf.io/reports/kubernetes-project-journey-report/
- https://kubernetes.io/docs/concepts/architecture/
- https://etcd.io/
- https://kubernetes.io/docs/concepts/workloads/pods/
- https://kubernetes.io/docs/reference/command-line-tools-reference/kubelet/
- Logos: https://commons.wikimedia.org/wiki/File:Google_Cloud_logo.svg, https://commons.wikimedia.org/wiki/File:IBM_Cloud_logo.png, https://commons.wikimedia.org/wiki/File:Amazon_Web_Services_Logo.svg, https://commons.wikimedia.org/wiki/File:AlibabaCloudLogo.svg, https://commons.wikimedia.org/wiki/File:Microsoft_Azure.svg

Sources

- https://kubernetes.io/docs/setup/best-practices/cluster-large/
- https://kubernetes.io/docs/setup/production-environment/tools/kubeadm/ha-topology/
- https://github.com/kubernetes/minikube/
- https://minikube.sigs.k8s.io/docs/
- https://kubernetes.io/docs/tasks/tools/#kubectl
- https://github.com/kubernetes/kubeadm
- https://kubernetes.io/docs/concepts/overview/working-with-objects/namespaces/